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TOURISM IN SAUDI ARABIA

INSIGHTS FROM THE PAST, TRENDS OF THE PRESENT, FORECASTS FOR THE FUTURE

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ABSTRACT:

This report provides a detailed analysis of tourism in Saudi Arabia, focusing on its trends, historical data, and future projections. The study uses time series analysis to forecast tourist arrivals between 2015 and 2025. Key factors influencing tourism in Saudi Arabia, including socio-economic developments, infrastructure improvements, and global events like the World Cup, are explored. The findings show significant growth in the tourism sector, with projections indicating a steady increase in tourist numbers. This analysis aims to help policymakers and businesses understand future tourism trends and make informed decisions.

KEYWORDS:

Tourism, Saudi Arabia, Time Series Analysis, Forecasting, Tourism Trends, Future Projections, Economic Impact, Tourism Growth, Vision 2030, Inbound Tourism, Cultural Tourism, Economic Diversification, Pilgrimage Tourism, Tourism Infrastructure, Post-COVID Tourism.

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INTRODUCTION

Tourism in the Kingdom of Saudi Arabia has undergone a remarkable transformation in recent years. Traditionally, the country was primarily known for religious tourism, especially during the Hajj and Umrah seasons. However, the tourism sector is now expanding to embrace a broader range of activities, aligning with Saudi Arabia's broader Vision 2030 for economic diversification and cultural openness. The government has recognized the crucial role tourism can play in reducing dependence on oil revenues and fostering sustainable economic growth, leading to strategic investments and ambitious reforms aimed at positioning Saudi Arabia as a top global tourism destination.

With the launch of Vision 2030, Saudi Arabia has embarked on a comprehensive journey to reshape its global image and attract millions of international visitors. This national strategy highlights the Kingdom's rich cultural heritage, historical landmarks, stunning natural landscapes, and diverse entertainment offerings. From the picturesque Red Sea coastline to UNESCO World Heritage sites like Al-Ula and Diriyah, Saudi Arabia is revealing its hidden treasures to both domestic and international tourists. Major projects such as NEOM, The Red Sea Project, Qiddiya, and Amaala reflect the country's commitment to creating world-class tourism experiences that cater to a variety of interests, including adventure tourism, wellness, and luxury travel.

In recent years, the sector has gained significant momentum, particularly with the introduction of the tourist visa in 2019. This initiative allowed travelers from multiple countries to visit Saudi Arabia more easily, thereby boosting international accessibility. The opening of the sector has also led to considerable infrastructure development, including airports, hotels, transportation networks, and entertainment facilities. Furthermore, the Kingdom has promoted festivals, concerts, sporting events, and cultural exhibitions, garnering both regional and global attention.

The transformation of tourism in Saudi Arabia is not solely an economic shift; it also has substantial social and cultural implications. The opening of the tourism sector has fostered cultural exchange and increased public engagement with national heritage. This shift has contributed to reshaping societal perceptions, particularly in regard to the role of women in the workforce and public life.

Moreover, the rapid growth of tourism has created thousands of job opportunities, empowering young Saudis to actively contribute to the development of this dynamic industry.

Saudi Arabia is undergoing a historic shift in its tourism landscape. Moving beyond its traditional focus on religious tourism, the Kingdom is embracing a more diverse and competitive tourism sector. With robust government support, visionary planning, significant infrastructure investment, and active international promotion, Saudi Arabia is emerging as a rising star in the global tourism industry, poised to welcome millions of visitors in the coming years.

1.1 TERMINOLOGY AND DEFINITIONS:

1. Religious Tourism

Refers to tourism centered around visiting sacred places and religious sites, such as Mecca and Medina, where millions of Muslims visit annually for Hajj and Umrah. This form of tourism is not only a spiritual journey but also has a significant impact on the Kingdom's economy. It attracts millions of international visitors each year, contributing to various sectors including hospitality, transportation, and retail. Additionally, the government has made substantial investments in expanding the infrastructure in Makkah and Medina to accommodate the growing number of pilgrims.

2. Ecotourism

A type of tourism focused on visiting natural areas with the purpose of learning about them, protecting the environment, and contributing to the preservation of natural resources.

3. Tourism Sector

Refers to the part of the economy that includes all activities and services related to tourism, such as hotels, airlines, transportation, and leisure activities.

4. Tourism Investment

Refers to the capital invested in developing tourism projects, such as building hotels, resorts, and other facilities aimed at attracting tourists.

5. Tourist Destinations

Locations that attract tourists for sightseeing, relaxation, or activities, such as Mecca, Al-Ula, and the Red Sea.

6. Tourism Festivals

Events organized to attract tourists, such as the Riyadh Winter Festival, Janadriyah Festival, or Red Sea International Film Festival.

7. Inbound Tourism (Tourism from Outside the Kingdom)

Refers to tourists traveling from other countries to visit Saudi Arabia. In this context, only visitors coming from outside the country are considered tourists when they travel to Saudi Arabia. For instance, if someone travels within Saudi Arabia from Riyadh to the Eastern Province for leisure purposes, they would not be counted as a tourist in the Eastern Province, as they are not an international visitor. In contrast, a foreign tourist traveling from their country to the Eastern Province for sightseeing or leisure would be considered an inbound tourist.

1.2 DATA SOURCE:

The primary data source for this project was obtained from the website datasaudi.sa, which provides monthly and yearly data on the number of tourists in Saudi Arabia from 2015 to 2023. This dataset served as the foundation for analyzing tourism trends within the Kingdom. Additionally, various statistical tools such as Excel and Power BI were utilized to process and visualize the data, ensuring a comprehensive analysis of the trends and patterns observed in the tourism sector. However, some missing or inaccurate data, specifically related to certain months and years, was retrieved from the General Authority for Statistics (GASTAT). After contacting the authority directly, they promptly provided the necessary and accurate data to fill in the gaps.

For the year 2024, no official data was available at the time of conducting this study. I reached out to both the General Authority for Statistics and the Ministry of Tourism, and both entities confirmed that the data for 2024 had not yet been released.

1.3 WHY RIYADH, EASTERN PROVINCE, AND MAKKAH?

In this study, we will focus specifically on three key regions in Saudi Arabia: **Riyadh**, **Eastern Province**, and **Makkah**, as they are central to the country's tourism industry. Each of these regions has unique characteristics that significantly influence inbound tourism and the overall tourist numbers in Saudi Arabia.

1. **Riyadh**: As the capital city of Saudi Arabia, Riyadh is not only a political and administrative hub but also a growing cultural and business center. The city's modern infrastructure, along with its rich history and cultural heritage, draws both domestic and international tourists. Key attractions include the National Museum, Kingdom Centre, and Diriyah, which is a UNESCO World Heritage Site. Riyadh's role as the center of government and business also means it hosts numerous international conferences, exhibitions, and events, contributing to a steady influx of tourists year-round. With Vision 2030 and its focus on diversifying the tourism sector, Riyadh is expected to see increased growth in both business and leisure tourism. The upcoming projects like the King Salman Park and the Riyadh Art initiative will further enhance the city's appeal to international visitors, positioning it as a hub for cultural and recreational tourism.

2. **Eastern Province:** The Eastern Province, known for its oil-rich resources, is home to significant economic hubs like Dammam, Al Khobar, and Dhahran. The region's beautiful coastal cities along the Arabian Gulf, such as Al Khobar and Half Moon Bay, attract tourists seeking both leisure and business opportunities. The Eastern Province also hosts important cultural and historical sites, including the archaeological sites in Al-Ahsa, as well as modern shopping centers and entertainment venues that appeal to a diverse range of tourists. Additionally, with major oil and gas conferences being held regularly, the region also attracts business travelers, further boosting its tourism numbers.

3. **Makkah**: Makkah is the spiritual heart of Saudi Arabia and one of the most significant pilgrimage destinations in the world. It is home to the Masjid al-Haram and the Kaaba, making it a central destination for millions of Muslim tourists who visit annually for Hajj and Umrah. Beyond religious tourism, Makkah is increasingly focusing on its infrastructure and tourism offerings for non-pilgrims. The city's tourism sector has been diversifying with new developments such as luxury hotels, shopping malls, and entertainment facilities, which attract visitors year-round. Furthermore, the ongoing investments in Makkah's tourism infrastructure are expected to enhance its appeal to a broader range of international visitors, particularly as the Kingdom continues to focus on expanding its tourism sector under Vision 2030.

1.4 KEY VARIABLES FOR TOURISM ANALYSIS

In this study, the following key variables were employed to analyze tourist numbers across different regions of Saudi Arabia over time:

1. Tourist Numbers

• **Definition**: This variable represents the total number of tourists visiting Saudi Arabia, categorized by month and year, from 2015 to 2023, with projections for 2024 and 2025.

• **Source**: Data for tourist numbers were sourced from datasaudi.sa, with missing or inaccurate data supplemented through a request to the Saudi General Authority for Statistics, which provided the necessary and accurate data. To address missing or inaccurate data, statistical imputation methods such as moving averages and regression models were employed to ensure continuity and minimize bias.

• **Impact**: As the primary variable of focus, it enables the analysis of tourist inflows and serves as the foundation for forecasting future trends in tourism. Projections for 2024 and 2025 were generated using time series forecasting models, incorporating historical data trends to estimate future tourist numbers.

2. Time Period (Year/Month)

• **Definition**: This variable captures the temporal aspect of the data, broken down by year and month, covering the period from 2015 to 2023, with projections for 2024 and 2025.

• **Source**: Data was sourced from datasaudi.sa, with supplementary data from the General Authority for Statistics.

• **Impact**: Time is a critical factor for identifying seasonal trends and understanding fluctuations in tourist numbers across different periods.

3. Region (Riyadh, Eastern Province, Makkah)

• **Definition**: This variable reflects the geographical distribution of tourists, focusing on three key regions: Riyadh, the Eastern Province, and Makkah.

• **Source**: Data was obtained from datasaudi.sa and the Saudi General Authority for Statistics.

• **Impact**: By focusing on these regions, the study highlights regional differences in tourist inflows, which can be influenced by various factors such as economic activity, religious significance, and the attractiveness of regional landmarks.

1.5 RATIONALE OF THE STUDY

This study's rationale arises from the growing ignificance of tourism in Saudi Arabia, especially with the strategic focus outlined in Vision 2030, which aims to diversify the economy away from oil dependency. As Saudi Arabia aims to diversify its economy and reduce dependence on oil revenues, the tourism sector has become a major focus for economic growth and development.

In recent years, the Kingdom has witnessed a significant increase in both domestic and international tourism, with key regions such as Riyadh, Makkah, and the Eastern Province emerging as central hubs for visitors. The rising number of tourists, particularly during peak seasons such as Hajj and Ramadan, highlights the importance of understanding tourist inflows in these regions.

However, there is a gap in the available data concerning accurate forecasts of tourist numbers in these specific areas, which poses a challenge for strategic planning. This study addresses this gap by providing detailed analysis and forecasts for tourist arrivals in Riyadh, Makkah, and the Eastern Province from 2015 to 2025.

The findings of this research will provide policymakers and tourism planners with the necessary tools to make informed decisions based on reliable data, contributing to shaping future tourism strategies and supporting sustainable development in these regions. Furthermore, the study will enhance the broader understanding of tourism dynamics in Saudi Arabia, aligning with the country's strategic vision to position itself as a leading global tourism destination.

1.6 OBJECTIVES OF THE STUDY

The main objectives of this study are as follows:

1. To **analyze the trends in tourist** arrivals to Riyadh, Makkah, and the Eastern Province from 2015 to 2023, and to forecast tourist numbers for 2024 and 2025.

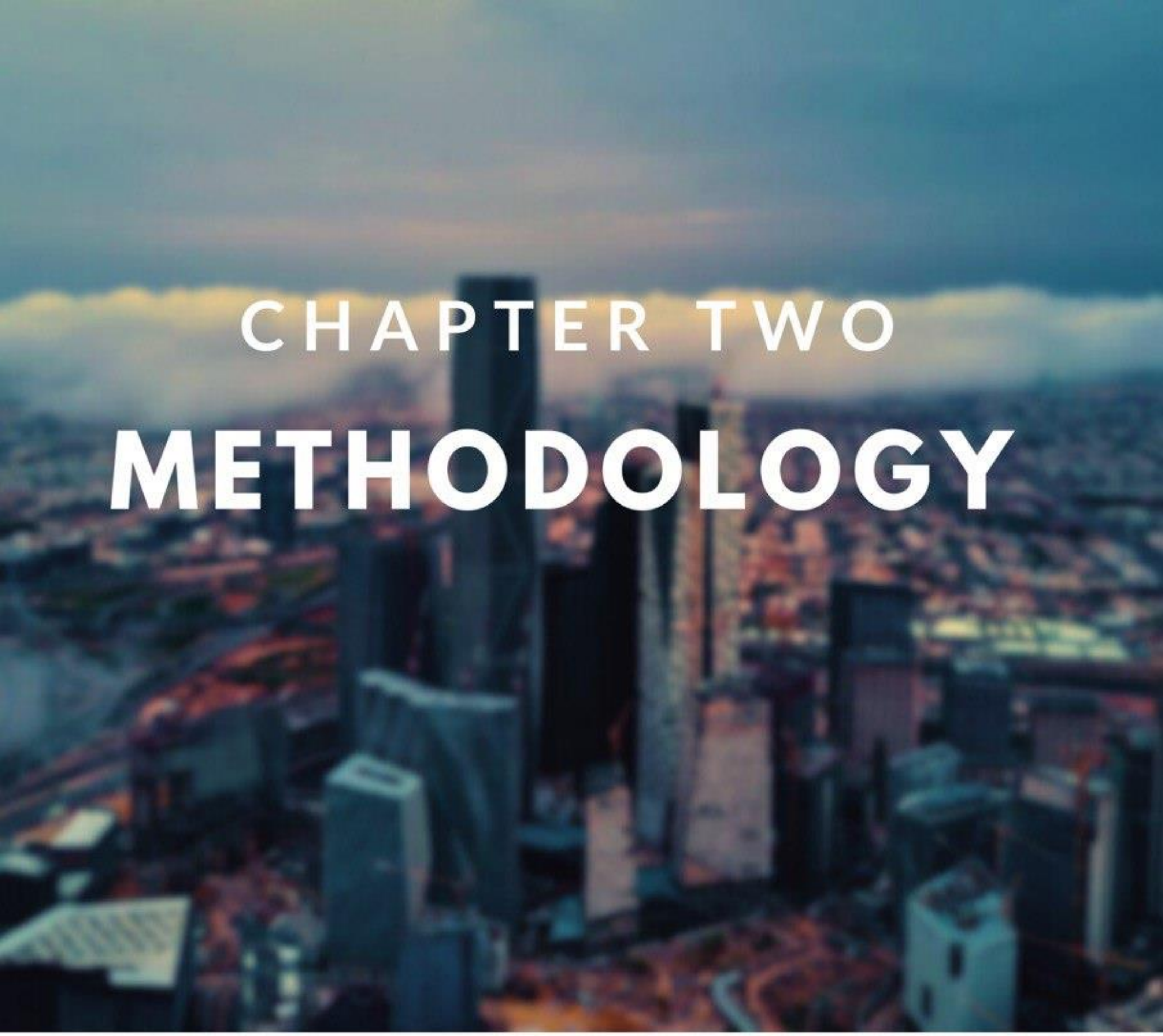
2. To **identify seasonal patterns** and provide accurate projections for tourist numbers during key periods such as Hajj and Ramadan, which influence regional tourism dynamics.

3. To **explore the factors influencing tourist numbers** in Riyadh, Makkah, and the Eastern Province, with particular focus on economic conditions, religious significance, and key regional attractions, while considering their impact on tourism trends**.**

4. To **provide insights for policymakers and tourism planners** by offering accurate forecasts and trends for tourism in these regions, aiding in the development of effective tourism strategies.

5. To **contribute to the literature on tourism dynamics in Saudi Arabia**, providing a foundation for future studies on tourism forecasting and regional analysis in the Kingdom.

6. To **estimate missing data** for 2024, as there were gaps in the available tourist numbers for that year. This study aims to provide estimated figures for 2024, contributing to the availability of complete data for future analysis and decision-making.



CHAPTER TWO

METHODOLOGY

THIS CHAPTER INCLUDES:

THIS CHAPTER PROVIDES A DETAILED OVERVIEW OF THE METHODOLOGY EMPLOYED IN THIS STUDY. IT EXPLAINS THE FUNDAMENTAL CONCEPTS, METHODS, AND TOOLS USED TO ANALYZE THE TOURIST DATA , THIS CHAPTER INCLUDES:

1. DEFINITION OF TIME SERIES: AN EXPLANATION OF WHAT TIME SERIES DATA IS, ITS CHARACTERISTICS, AND WHY IT IS USED TO ANALYZE TRENDS AND PATTERNS OVER TIME.

2. TREND ANALYSIS: A DESCRIPTION OF WHAT A TREND IS IN THE CONTEXT OF TIME SERIES ANALYSIS AND HOW IT CAN BE IDENTIFIED IN THE DATA.

3. SEASONALITY: AN EXPLANATION OF SEASONAL VARIATIONS, PARTICULARLY FOCUSING ON HOW THEY IMPACT TOURISM PATTERNS, INCLUDING THE ROLE OF KEY PERIODS SUCH AS HAJJ AND RAMADAN.

4. WHITE NOISE: AN INTRODUCTION TO THE CONCEPT OF WHITE NOISE IN TIME SERIES DATA AND HOW IT AFFECTS MODEL ACCURACY AND FORECASTING.

5. STATIONARITY: A DISCUSSION OF WHAT STATIONARITY MEANS IN TIME SERIES ANALYSIS, WHY IT IS IMPORTANT, AND HOW STATIONARITY IS TESTED.

6. DATA COLLECTION, SOURCES, AND TRANSFORMATION METHODS: A BRIEF OVERVIEW OF THE DATA SOURCES USED IN THE STUDY, FOLLOWED BY THE METHODS USED TO TRANSFORM AND PREPARE THE DATA FOR ANALYSIS, SUCH AS HANDLING MISSING VALUES, ADJUSTING FOR SEASONALITY, AND ENSURING DATA CONSISTENCY.

7. FORECASTING TECHNIQUES: A DESCRIPTION OF THE TECHNIQUES AND METHODS EMPLOYED TO FORECAST TOURIST ARRIVALS FOR THE YEARS 2024 AND 2025.

BY PROVIDING THESE FOUNDATIONAL EXPLANATIONS, THIS CHAPTER ENSURES A CLEAR UNDERSTANDING OF THE KEY CONCEPTS AND TOOLS USED IN THE ANALYSIS, SETTING THE STAGE FOR THE RESULTS AND CONCLUSIONS THAT FOLLOW IN THE LATER CHAPTERS.

TIME SERIES

A time series is a sequence of data points measured at successive time intervals, typically at uniform intervals. In this study, time series refers to the tourist data collected for Riyadh, Makkah, and the Eastern Province, measured monthly from 2015 to 2023. Time series data helps identify trends, seasonal patterns, and irregular fluctuations in the data.

2.1 TREND

A trend in a time series refers to a long-term increase or decrease in the data over time. This can reflect underlying shifts in the data that occur consistently over a period, such as an upward trend in tourist arrivals due to improved infrastructure or economic factors. Identifying trends is important for understanding how the data behaves over a long horizon.

2.2 SEASONALITY

Seasonality refers to regular and predictable changes in data that recur over a specific period, such as a year or a season. In tourism, seasonality often correlates with specific events, holidays, or weather patterns. For example, tourist arrivals might peak during the Hajj season or Ramadan. Understanding seasonality is key to making accurate predictions.

2.3 STATIONARITY

A time series is considered stationary when its statistical properties, such as mean, variance, and autocorrelation, do not change over time.This means that the series does not show trends or seasonal effects, and its behavior is consistent. If your time series data is non-stationary, transformations such as differencing or logarithms may be applied to make it stationary.

2.4 DIAGNOSTIC TESTS FOR STATIONARITY AND RESIDUAL PROPERTIES

This section outlines the statistical tests applied to assess the stationarity of the time series and to examine the variance behavior of residuals prior to model development.

2.4.1 ADF (Augmented Dickey-Fuller Test)

The Augmented Dickey-Fuller (ADF) test is a statistical test used to determine whether a time series is stationary or contains a unit root. Stationarity implies that the statistical properties of the series, such as mean and variance, remain constant over time. In contrast, the presence of a unit root indicates non-stationarity, often due to a stochastic trend, and suggests that differencing or transformation may be required to stabilize the series.

The hypotheses for the ADF test are formulated as follows:

- **Null hypothesis (H_0)**: The time series has a unit root (i.e., it is non-stationary).
- **Alternative hypothesis (H_1):** The time series does not have a unit root (i.e., it is stationary).

To interpret the results, the p-value obtained from the test is compared to a chosen significance level (commonly 0.05):

- If p-value < 0.05, we reject the null hypothesis, concluding that the series is stationary.
- If p-value ≥ 0.05, we fail to reject the null hypothesis, suggesting that the series is non-stationary and may require differencing or transformation.

2.4.2 KPSS (Kwiatkowski-Phillips-Schmidt-Shin) Test

The KPSS test is a statistical test used to assess the stationarity of a time series, but with a different hypothesis framework compared to the ADF test. While the ADF test assumes the presence of a unit root under the null hypothesis, the KPSS test assumes stationarity as the null.

The hypotheses for the KPSS test are as follows:

- **Null hypothesis (H_0):** The time series is stationary (around a mean or a deterministic trend).
- **Alternative hypothesis (H_1):** The time series is non-stationary (i.e., it contains a unit root).

Interpretation is typically based on the p-value:

- If p-value < 0.05, we reject the null hypothesis, suggesting that the series is non-stationary.
- If p-value ≥ 0.05, we fail to reject the null hypothesis, implying that the series is stationary.

2.4.3 ARCH (Autoregressive Conditional Heteroskedasticity) Test

The ARCH test is used to detect the presence of conditional heteroskedasticity in the residuals of a time series model. In other words, it checks whether the variance of the errors is constant over time (homoskedastic) or varies depending on past values (heteroskedastic).

The hypotheses of the ARCH test are:

- **Null hypothesis (H_0):** There is no ARCH effect (the residual variance is constant over time).
- **Alternative hypothesis (H_1):** There is an ARCH effect (the residual variance changes over time).

Interpretation:

- If p-value < 0.05, we reject the null hypothesis, indicating the presence of heteroskedasticity in the residuals.
- If p-value ≥ 0.05, we fail to reject the null hypothesis, suggesting no significant ARCH effects and thus constant variance.

2.5 TRANSFORMATIONS AND DIFFERENCING

In this section, the transformations and differencing techniques applied to the time series are discussed. These methods were implemented to stabilize variance and achieve stationarity, which are crucial steps before applying forecasting models. The choice of transformations and differencing were guided by the results of the diagnostic tests performed earlier.

2.5.1 Differencing

Differencing is a technique used to transform a non-stationary time series into a stationary one by removing trends and seasonality. This method involves subtracting a previous observation from the current observation to eliminate systematic patterns such as linear trends or seasonal fluctuations. Achieving stationarity is essential for many time series models, including ARIMA and its seasonal variants.

The general form of differencing is defined as:

Differenced value$=y_t - y_{t-d}$

Where d represents the order of differencing. When d = 1, the process is known as first differencing, which is typically used to remove linear trends:

$\nabla y_t = y_t - y_{t-1}$

In cases where the time series exhibits both trend and seasonality, seasonal differencing may also be applied using a seasonal lag s:

$\nabla_s y_t = y_t - y_{t-s}$

In this study, differencing was applied as needed based on stationarity tests. Both regular and seasonal differencing were utilized to stabilize the mean and variance of the series, thereby enabling the application of appropriate time series forecasting models.

2.5.2 Box-Cox Transformation

The Box-Cox transformation is a family of power transformations used to stabilize variance and make the data more normally distributed. It is particularly useful when dealing with time series data that exhibits non-constant variance (heteroscedasticity) or data that is skewed. The transformation is defined as follows:

$$g(x) = \begin{cases} \dfrac{x^{\lambda} - 1}{\lambda}, \lambda \neq 0 \\ ln(x), \lambda = 0 \end{cases}$$

Where Y is the original data and λ is a parameter that controls the transformation. The value of λ is typically chosen based on the data, and it can be estimated using maximum likelihood estimation (MLE). Common values for λ include:

- **λ = 1**: No transformation.
- **λ = 0**: Log transformation.
- **λ = 0.5**: Square root transformation.
- **λ = -1**: Inverse transformation

2.5.3 Log Transformation

In some cases, the Box-Cox transformation did not provide satisfactory results in stabilizing variance and normalizing the data. As an alternative, the logarithmic transformation was applied, which is often effective in dealing with data that exhibits a right skew or non-constant variance. The log transformation is defined as:

$g(x) = \ln(x)$

However, it is important to note that the log transformation can sometimes lead to wider confidence intervals due to the compression of large values, which may introduce additional uncertainty in predictions. To address this issue and improve the model's accuracy, a shrinkage technique was applied, as described in this study, to reduce the impact of these expanded intervals and ensure more reliable forecasts.

2.6 KALMAN FILTER FOR MISSING DATA IMPUTATION

In this study, the Kalman filter was employed to address the issue of missing data, particularly for the year 2024. The Kalman filter is an efficient recursive method for estimating the state of a linear dynamic system from noisy observations. It operates by predicting the next state of the system, then correcting this prediction based on the observed data. This approach is especially useful when dealing with time series data that may contain missing or incomplete values.

Rather than relying on typical imputation techniques or using automatic models like AutoARIMA, I opted to design a custom forecasting model. This allowed for more control over the assumptions and conditions applied to the data. The Kalman filter method ensured that the missing data points were appropriately estimated based on available observations, allowing for a more accurate and coherent analysis. This technique was carefully integrated with the analysis process, ensuring that all model assumptions and diagnostic checks were met, similar to the process of validating filter conditions.

By using a custom model and employing the Kalman filter for missing data, I was able to maintain the integrity of the time series data and proceed with a more tailored approach to the analysis, while ensuring compliance with statistical assumptions.

2.7 White Noise

White noise refers to a sequence of random data points that exhibit a constant mean and variance, with no correlation between data points over time. In the context of time series, white noise is often seen as random “background noise” that lacks any predictable pattern or trend. It represents the ideal scenario for the residuals in time series models, as it indicates that the model has successfully captured all systematic patterns in the data, leaving only random fluctuations.

Conditions for White Noise:

1. **Zero Mean:** The data should have a mean of zero, implying that the values are symmetrically distributed around zero.

2. **Constant Variance:** The variance of the data should remain constant over time, ensuring that the fluctuations are consistent across the series.

3. **No Autocorrelation:** The data points should not exhibit any correlation with each other over time. In other words, the value at time t should have no relationship with the value at time t-1 or any other time point.

4. **Randomness:** The values should be completely random, with no predictable pattern, trend, or seasonality.

2.8 Autocorrelation Function (ACF) :

The Autocorrelation Function (ACF) is a fundamental statistical tool used in time series analysis to quantify the degree of similarity between a time series and a lagged version of itself over successive time intervals. Specifically, it measures the linear relationship between current and past values across different lag periods. By calculating autocorrelation at lag k, the ACF reflects how strongly a value at time t is correlated with a value at time t - k.

ACF is essential in diagnosing the presence of autocorrelation patterns such as seasonality, cyclicity, or persistence within the data. A slow decay in the ACF plot often indicates non-stationarity, suggesting that the series may require differencing to stabilize the mean. Alternatively, the presence of spikes at seasonal lags (e.g., at lag 12 for monthly data) may indicate seasonality.

In model identification, the ACF plot is particularly useful in determining the order q of the Moving Average (MA) component within ARIMA models. For example, if the ACF plot exhibits a sharp cut-off after lag q and is near zero beyond this point, it suggests the presence of an MA(q) structure.

Mathematically, the ACF at lag k is defined as: $\frac{\operatorname{cov}(Y_t, Y_{t-1})}{\operatorname{var}(Y_t)}$

where Cov (y_t, y_{t-k}) is the covariance between the observations at time t and t - k, and σ^2 is the variance of the series.

2.9 PARTIAL AUTOCORRELATION FUNCTION (PACF):

The Partial Autocorrelation Function (PACF) extends the concept of autocorrelation by isolating the direct effect of a specific lag on the current observation, after accounting for the contributions of all shorter lags. While ACF captures both direct and indirect correlations (e.g., the influence of lag 1 may propagate to lag 2 and beyond), the PACF isolates the pure or partial correlation at lag k by eliminating intermediate effects.

The PACF is especially critical when identifying the order p of the AutoRegressive (AR) component in ARIMA models. A PACF plot that shows significant spikes up to lag p, followed by insignificant values, typically suggests an AR(p) structure.

For instance, if the PACF displays significant values at lags 1 and 2 but cuts off afterward, this indicates a potential AR(2) model. In contrast to ACF, which may decay gradually due to the cumulative influence of past observations, PACF emphasizes only the direct relationships.

Mathematically, PACF values are computed through recursive regression methods, where at each lag k, the correlation is adjusted for the linear effect of all lags less than k.

Both ACF and PACF are visualized through correlograms, where significant spikes outside the confidence bands (typically 95%) suggest noteworthy lags that should be considered in model specification.

Important Methodological Note:

In time series analysis, the selection of the number of parameters (such as the value of p in a SARIMA model) is a key factor that significantly affects model performance. It is well known that increasing the number of parameters can lead to greater model complexity, which may result in issues like overfitting or poor performance in future predictions. In this context, it is advisable to reduce the number of parameters when more complex models do not show significant improvements in forecasting accuracy.

For instance, as Hyndman and Athanasopoulos noted in their book Forecasting: Principles and Practice (2018), “reducing the number of parameters in the model can lead to significant improvements in performance if more complex models do not provide substantial improvements in accuracy measures like MAPE” (Hyndman & Athanasopoulos, 2018, p. 121). Therefore, focusing on simpler models that demonstrate better performance can be an effective approach to achieving more accurate forecasts.

This approach aligns with the practice in ARIMA and SARIMA models where reducing the number of parameters is allowed based on the actual model performance rather than theoretical criteria alone.

2.10 TIME SERIES

A time series is a sequence of data points collected or recorded at specific time intervals, typically equally spaced, such as daily, monthly, quarterly, or yearly. The key characteristic of a time series is that the observations are not independent of each other, as they follow a chronological order. This temporal ordering introduces patterns such as trends, seasonality, cycles, and random fluctuations that differentiate time series data from cross-sectional data.

In many real-world applications, time series analysis is used to understand the underlying structure and dynamics of the data over time, as well as to make forecasts. Fields such as economics, finance, environmental studies, healthcare, and tourism frequently rely on time series data for informed decision-making.

Key Characteristics of Time Series Data:

1. **Trend**: A long-term increase or decrease in the data values over time.
2. **Seasonality:** Repeated patterns or cycles observed at regular intervals (e.g., higher tourist arrivals during summer or religious holidays).
3. **Cyclicality:** Fluctuations that occur over longer periods, often influenced by economic or social factors, but without a fixed period like seasonality.
4. **Irregular/Random Component:** Unpredictable variations or noise that cannot be attributed to trend, seasonality, or cycles.

Importance of Time Series Analysis:

Time series analysis allows researchers and practitioners to:

- Identify underlying patterns and components (trend, seasonality, etc.).
- Model the behavior of a system or process over time.
- Forecast future values based on historical data.
- Detect anomalies or outliers within the data.
- Improve planning and decision-making in dynamic environments.

2.10.1 Autoregressive Model (AR):

The Autoregressive (AR) model is one of the foundational models in time series forecasting. It assumes that the current value of the series can be explained by a linear combination of its previous values, plus a stochastic error term. In other words, the AR model suggests that past observations have a direct influence on the present value of the series.

Mathematically, an AR model of order p, denoted as AR(p), is defined as:

$$y_t = \varepsilon_t + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p}$$

Where:

- y_t is the current value of the time series at time t.
- ϕ_1, ϕ_2, ϕ_p are the autoregressive coefficients.
- ε_t is a white noise error term.

In simple terms, the AR model captures how much the current value depends on its p past values. This model is especially useful when there is evidence of persistence or momentum in the series.

ACF and PACF behavior for AR model:

- **PACF**: The PACF of an AR(p) model will show significant spikes up to lag p, followed by a sharp cut-off to approximately zero afterward. This is because PACF isolates the direct relationship between the current value and its pth lag while removing the influence of intermediate lags.
- **ACF**: The ACF of an AR model usually decays gradually or exponentially with increasing lags, reflecting the lingering influence of past values.

2.10.2 Moving Average Model (MA):

The Moving Average (MA) model is another classical time series model. Unlike the AR model, which uses past observations, the MA model expresses the current value as a linear function of past error terms (also known as white noise or shocks).

Mathematically, an MA model of order q, denoted as MA(q), is defined as:

$$y_t = \varepsilon_t - \theta_1\varepsilon_{t-1} - \theta_2\varepsilon_{t-2} - \cdots - \theta_q\varepsilon_{t-q}$$

Where:

• $\theta_1, \theta_2, \ldots, \theta_q$ are the moving average coefficients.

• ε_t is the white noise error term at time t. The MA model captures short-term dependencies based on past random shocks. It is particularly useful for series where random disturbances are influencing the values but without a clear autoregressive structure.

ACF and PACF behavior for MA model:

• **ACF**: The ACF of an MA(q) model will display significant spikes up to lag q, followed by a sharp cut-off to approximately zero beyond lag q.

• **PACF**: The PACF for an MA model typically decays gradually or exponentially as the lag increases, similar to how ACF behaves for an AR model but reversed in terms of where the cut-off occurs.

2.10.3 Autoregressive Moving Average Model (ARMA):

The ARMA model combines both the AR and MA components to explain the behavior of a stationary time series. The model incorporates both the autoregressive dependence on past values and the moving average dependence on past error terms.

An ARMA(p, q) model is represented as:

$$y_t = \phi_0 + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \ldots + \phi_p y_{t-p} + \theta_1\varepsilon_{t-1} + \theta_2\varepsilon_{t-2} + \ldots + \theta_q\varepsilon_{t-q} + \varepsilon_t$$

This hybrid structure makes ARMA models powerful for modeling stationary data where both past observations and past shocks contribute to future values.

ACF and PACF behavior for ARMA model:

• **ACF**: The ACF of an ARMA model typically decays gradually (i.e., neither cuts off at lag p nor q but rather tails off), depending on the values of p and q.

• **PACF**: Similarly, the PACF will also display a gradual decay.

Because ARMA models integrate both AR and MA characteristics, their correlograms tend to show more complex decaying patterns compared to pure AR or MA models.

2.10.4 Integrated Autoregressive Moving Average (ARIMA) Model:

The ARIMA model, which stands for AutoRegressive Integrated Moving Average, is a generalization of ARMA used for non-stationary time series. It adds a differencing step to the ARMA framework to make the series stationary before applying AR and MA components.

The ARIMA model is denoted as ARIMA(p, d, q) where:

• p is the order of the AR term.

• d is the number of non-seasonal differences needed to achieve stationarity.

• q is the order of the MA term.

The differencing component (d) addresses non-stationarity by subtracting the previous value from the current value (or by performing higher-order differencing).

Mathematically, for a first difference (d = 1), the model is applied to (y_t - y_{t-1}), and then ARMA(p, q) is fitted on the differenced series.

In time series forecasting, ARIMA is a highly flexible model that can model a wide variety of real-world data exhibiting trends or other forms of non-stationarity.

ACF and PACF behavior for ARIMA model:

• For ARIMA models, the ACF and PACF should be evaluated **after differencing the series (if d > 0)**, as the plots on the raw data may not provide clear insights.

• Once the series is stationary:

• If the model behaves like an **ARIMA(p, 0, 0)** (i.e., an AR model on the differenced series), expect PACF to cut off after p lags and ACF to decay.

• If it behaves like **ARIMA(0, 0, q)** (i.e., MA on the differenced series), expect ACF to cut off after q lags and PACF to decay.

• In general, if both AR and MA components are present, both ACF and PACF will exhibit slow decays.

2.10.5 Seasonal Autoregressive Integrated Moving Average (SARIMA)

The Seasonal Autoregressive Integrated Moving Average (SARIMA) model is an extension of the ARIMA model designed to handle time series data that exhibits both non-seasonal and seasonal patterns. While ARIMA is effective for non-seasonal data that becomes stationary after differencing, SARIMA incorporates additional seasonal components to account for recurring seasonal behavior in the series.

The SARIMA model is denoted as: SARIMA(p, d, q) * (P, D, Q)s

Where:

• p, d, q are the non-seasonal autoregressive, differencing, and moving average orders (as in ARIMA).

• P, D, Q are the seasonal autoregressive, seasonal differencing, and seasonal moving average orders.

• s is the length of the seasonal cycle (e.g., s = 12 for monthly data with yearly seasonality).

SARIMA Model Equation:

The general SARIMA equation combines both seasonal and non-seasonal components and is written as:

$$\Phi_P(B^s)\phi_p(B)(1-B)^d(1-B^s)^D y_t = \Theta_Q(B^s)\theta_q(B)\epsilon_t$$

Where:

• **B** is the backshift operator.

• $(1 - B^s)^D$: represents seasonal differencing to remove seasonal trends.

• $\Phi_P(B^s)$ and $\Theta_Q(B^s)$ represent the seasonal AR and MA terms.

• $\phi_p(B)$ and $\theta_q(B)$ represent the non-seasonal AR and MA terms.

When to Use SARIMA?

SARIMA is particularly useful when the data shows:

• Regular seasonal fluctuations (e.g., monthly tourist numbers peaking in certain months).

• A trend combined with seasonality.

• Non-stationary behavior that requires both differencing and seasonal differencing to achieve stationarity.

For example, in the context of tourism data, visitor counts may increase during specific months each year (e.g., during religious holidays or school vacations), which would be modeled effectively using SARIMA.

Differences Between ARIMA and SARIMA:

Feature	ARIMA	SARIMA
Seasonality	Does not explicitly model seasonality.	Specifically includes seasonal components.
Notation	ARIMA(p, d, q)	SARIMA(p, d, q) × (P, D, Q)s
Differencing	Only non-seasonal differencing (d).	Uses both non-seasonal (d) and seasonal (D) differencing.
Use case	Suitable for non-seasonal or weakly seasonal data.	Best suited for data with strong seasonal patterns.
Components	AR and MA terms only for non-seasonal lags.	AR and MA terms for both seasonal and non-seasonal lags.
Complexity	Less complex.	More complex due to additional seasonal terms.

2.10.6 SARIMAX Model:

In this study, due to significant changes in the data caused by external factors such as the COVID-19 pandemic, the SARIMAX (Seasonal Autoregressive Integrated Moving Average with Exogenous variables) model was utilized instead of SARIMA. The SARIMAX model extends SARIMA by including exogenous variables (i.e., external factors) that may influence the time series data, such as changes in tourist behavior during the pandemic. This adjustment was made to account for the impact of these external variables, providing a more accurate representation of the data during this period of disruption.

Interpretation of ACF and PACF for SARIMAX Model:

When applying SARIMAX, the ACF and PACF plots help identify both seasonal and non-seasonal parameters:

• Seasonal ACF and PACF: Significant spikes at lags that are multiples of the seasonal period (e.g., lag 12, 24 for monthly data with annual seasonality).

• Non-seasonal ACF and PACF: Similar behavior to ARIMA models but with additional seasonal patterns layered on top.

For example:

• A spike at lag 12 in ACF suggests a seasonal MA(1) component (i.e., Q = 1).

• A spike at lag 12 in PACF suggests a seasonal AR(1) component (i.e., P = 1).

2.11 MODEL DIAGNOSTICS (CHECKING DIAGNOSTIC)

After fitting a time series model such as SARIMA, it is essential to perform model diagnostics to assess whether the model adequately captures the structure of the data and to verify that the residuals behave as white noise. Model diagnostics help identify if any patterns remain unexplained, which may suggest the need for model refinement.

Key Steps in Model Diagnostics:

1. Residual Analysis:

The residuals (errors) are the differences between the observed values and the model's fitted values. For a well-fitted model, residuals should behave like white noise, meaning they should:

• Have a mean close to zero.

• Show constant variance (homoscedasticity).

• Be uncorrelated over time (no significant autocorrelation).

• Follow a normal distribution (approximately, especially for forecasting intervals).

2. ACF and PACF of Residuals:

• The ACF plot of residuals is inspected to check for any significant autocorrelations remaining. All autocorrelations should lie within the 95% confidence interval bands.

• The PACF plot of residuals serves a similar purpose by identifying partial correlations that might remain unexplained.

If significant spikes are observed, it suggests that the model may not fully capture the temporal dependencies, and further adjustments to the model parameters might be necessary.

3. Ljung-Box Test (or Box-Pierce Test):

A statistical test used to determine whether the residuals are independently distributed. The null hypothesis states that residuals exhibit no autocorrelation up to a certain lag.

• p-value > 0.05 indicates no significant autocorrelation (model is adequate).

• p-value < 0.05 suggests residual autocorrelation (model may require refinement).

4. Normality of Residuals:

• A histogram or a Q-Q plot is typically used to assess if the residuals are approximately normally distributed.

• Non-normal residuals might impact confidence intervals but do not always invalidate the model, especially in large samples.

5. Residual Variance Plot (Homoscedasticity Check):

A plot of residuals versus fitted values is used to check for any patterns or changing variance (heteroscedasticity). A good model will show residuals scattered randomly without patterns or funnel shapes.

Why Diagnostics are Important:

Model diagnostics are critical to ensure:

- The model does not overfit or underfit the data.
- The model is reliable for forecasting future values.
- The assumptions of the time series model are reasonably satisfied.

Failing to conduct thorough diagnostics may lead to inaccurate forecasts and misleading conclusions about the data patterns.

In this project, after fitting the SARIMA model to the tourism data, several diagnostic checks were performed. The ACF and PACF of residuals indicated no significant autocorrelation, and the Ljung-Box test returned a p-value greater than 0.05, suggesting that the residuals behave as white noise. Additionally, the residuals were approximately normally distributed, and no signs of heteroscedasticity were detected, confirming that the model is a good fit for the data and suitable for forecasting purposes.

2.12 FORECASTING

Forecasting is the process of using a trained model to predict future values based on historical data. In this project, the SARIMA model was used to forecast tourist numbers in Saudi Arabia for the upcoming years (2024 and 2025). Once the model was trained and forecasts were generated, the accuracy of these predictions was assessed using various metrics to evaluate how well the model predicts future values.

Future Forecasts

After fitting the SARIMA model to the historical data, forecasts were made for the future years, particularly focusing on regions like Riyadh, Makkah, and the Eastern Province. These forecasts were used for planning purposes, such as predicting tourist numbers during specific periods of the year, like the Hajj season or official holidays.

Forecast Accuracy Evaluation

The accuracy of the forecasts is crucial to check how well the predicted values align with the actual observed values. There are several ways to evaluate forecast accuracy, with the most commonly used metrics being **MAPE** (Mean Absolute Percentage Error) and **MSRE** (Mean Squared Relative Error).

1. **MAPE (Mean Absolute Percentage Error)**

   MAPE is used to measure the average percentage of absolute errors between the predicted and actual values. It is one of the most popular metrics for forecast accuracy, and is calculated as follows:

$$MAPE = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{\widehat{y_i} - y_i}{y_i}\right| \times 100$$

   Where:
   $\widehat{y_i}$ = Predicted value for the i^{th} data point
   y_i= Actual value for the i^{th} data point
   n = number of observations

   In this study, MAPE was calculated using the cross-validation approach, rather than relying on a single train-test split. Cross-validation is a resampling method used to assess the generalizability and robustness of predictive models. Specifically, the time series data was divided into multiple training and validation sets in a rolling or expanding window fashion, ensuring that the temporal structure of the data is preserved. The model was trained on each training subset and evaluated on the corresponding validation subset, and the MAPE values obtained from each iteration were then averaged to obtain a more reliable estimate of forecast accuracy.

Advantages of MAPE:

- **Easy to understand and interpret**, as it expresses the error as a percentage.
- Useful for **comparing forecasts** across different datasets.
- When used with cross-validation, it provides a **more stable and realistic evaluation** of model performance over time.

2. **MSRE (Mean Squared Relative Error)**

MSRE is another metric used to measure the forecast accuracy, focusing on the relative error squared. It is defined as:

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(\hat{y}_i - y_i)^2}$$

Where:
$\hat{y}_i$ = Predicted value for the ith data point
y_i= Actual value for the ith data point
n = number of observations

Advantages of MSRE:

• Emphasizes larger errors by squaring the relative errors, making it more sensitive to significant forecasting mistakes.

• Helps in identifying larger discrepancies between predicted and actual values.

Backtesting in Forecasting Models:

In time series analysis, backtesting is a crucial process for evaluating the performance of forecasting models by using historical data that was not used during model training. This procedure involves using previous data from specific time periods to test how well the model can predict future outcomes. Through backtesting, the model's accuracy is evaluated under real-world conditions that it might have faced in the past.

As noted by Hyndman and Athanasopoulos (2018) in their book Forecasting: principles and practice, backtesting is a useful tool for assessing the performance of forecasting models, where models are tested on data they have not been trained on previously. This allows for an assessment of how well the model provides predictions based on the historical data available.

Backtesting in this Context:

In the context of my project on forecasting tourism numbers in Saudi Arabia for 2025, 2024 data was treated as forecasted, not actual data, meaning it could not be directly used to test the model's quality. Therefore, as outlined in the literature (Hyndman & Athanasopoulos, 2018), I conducted the backtest using the actual data for 2023, making this process a backtest where the model's accuracy was tested using historical data that had not been previously used in the model.

In this case, it was impossible to test the model with the 2024 data because it was forecasted and not actually collected. Therefore, testing the model with the 2023 data provided a realistic evaluation of the model’s performance in forecasting future results, allowing me to assess its accuracy using MAPE (Mean Absolute Percentage Error) calculated during the backtesting process.

CHAPTER THREE

STATISTICAL ANALYSIS AND FORECASTING

THIS CHAPTER INCLUDES:

1. A COMPREHENSIVE STATISTICAL ANALYSIS OF THE DATA, INCLUDING DESCRIPTIVE STATISTICS (MEAN, MEDIAN, STANDARD DEVIATION, ETC.).
2. EXAMINATION AND TRANSFORMATION OF THE DATA TO ENSURE STATIONARITY AND SUITABILITY FOR MODELING.
3. AN EXPLANATION OF THE CHOSEN FORECASTING MODEL (SARIMA) AND ITS APPLICATION TO THE DATA.
4. DETAILED EVALUATION OF MODEL PERFORMANCE USING ERROR METRICS SUCH AS MAPE AND MSRE.
5. ANALYSIS OF THE RESIDUALS AND THEIR IMPLICATIONS FOR MODEL ACCURACY.
6. FORECASTING OF FUTURE DATA POINTS FOR THE YEARS 2024 AND 2025.
7. A COMPARISON OF PREDICTED VALUES WITH HISTORICAL TRENDS AND PATTERNS.
8. DISCUSSION OF THE RESULTS IN RELATION TO THE PROJECT'S OBJECTIVES AND THE IMPLICATIONS OF THE FORECASTS.
9. LIMITATIONS OF THE MODEL AND POTENTIAL AREAS FOR FUTURE RESEARCH OR MODEL IMPROVEMENT.
10. SUMMARY OF KEY FINDINGS AND THEIR RELEVANCE TO DECISION-MAKING OR POLICY IMPLICATIONS.

DESCRIPTIVE ANALYSIS

In this section, we will provide a descriptive analysis of the data collected for the number of tourists in the three cities: Riyadh, Makkah, and the Eastern Region of Saudi Arabia. Descriptive statistics help to summarize the key features of a dataset, offering insights into the central tendency, variability, and overall distribution of the data. This analysis will include key statistics such as the mean, median, standard deviation, range, and interquartile range (IQR), which will allow us to better understand the patterns and trends in tourism across the regions.

3.1 DESCRIPTIVE STATISTICS

The following table presents the descriptive statistics for the number of tourists in Riyadh, Makkah, the Eastern Region, and the total of these three regions from 2015 to 2023 :

Statistic	Riyadh	Makkah	Eastern Region	Total of Three Regions
Mean	179,744.82	744,694.14	264,830.99	1,189,269.96
Median	182,951.24	813,698.54	251,197.79	1,310,846.76
Variance	10,356,662,866	318,758,562,423	20,079,335,757	526,238,162,473
Standard Deviation	101,767.69	564,587.07	141,701.57	725,422.75
Min	11,000.96	965.60	4,343.33	16,309.89
Max	721,159.94	2,041,278.00	639,694.27	2,730,748.00
Range	710,158.98	2,040,312.40	635,350.94	2,714,438.11
IQR	119,695.08	969,395.57	162,914.69	1,083,330.84
Skewness	1.48	0.22	0.22	0.04
Kurtosis	6.29	-0.95	-0.27	-0.92

Interpretation:

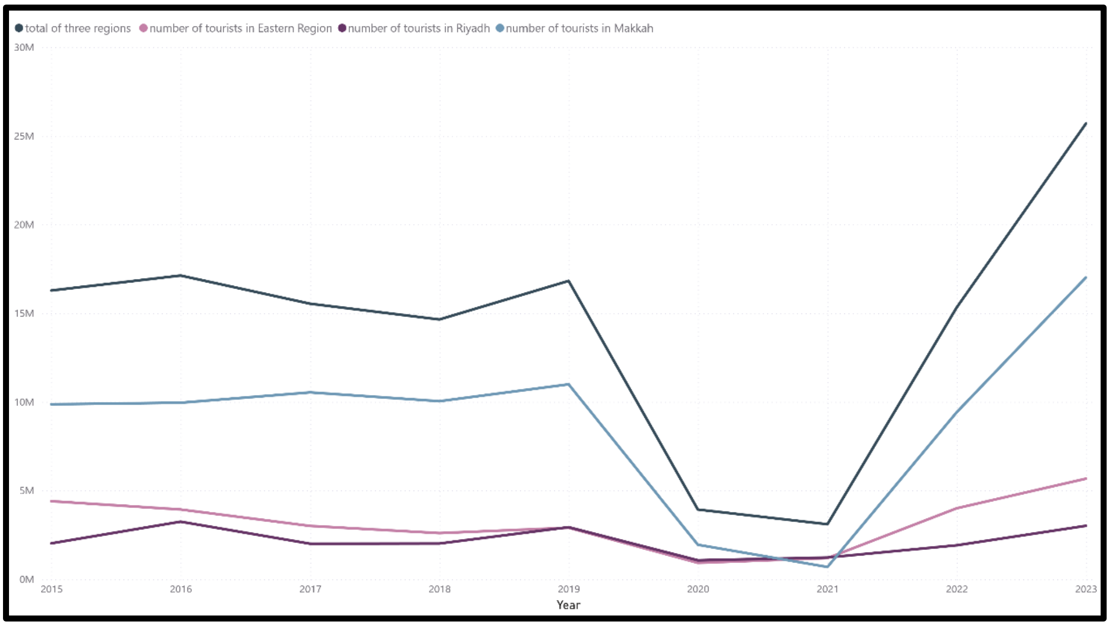
i. **Mean:** The average number of tourists in Riyadh is around 179,744, in Makkah is 744,694, and in the Eastern Region is 264,830. When combined, the total mean number of tourists across the three regions is approximately 1.19 million.

ii. **Median:** The median values show that the middle value for Riyadh is 182,951, for Makkah is 813,698, and for the Eastern Region is 251,197. The total combined median is 1.31 million.

iii. **Variance and Standard Deviation:** Makkah has the highest variance (and standard deviation), indicating greater variability in the number of tourists there compared to Riyadh and the Eastern Region. The total combined variance is quite high, reflecting the differences across regions.

iv. **Min and Max:** The minimum and maximum values show the lowest and highest tourist numbers. For example, Makkah has the highest maximum (2,041,278), while Riyadh has a lower minimum.
v. **Range:** The range is the difference between the max and min values. Makkah has the largest range, followed by Riyadh and the Eastern Region.
vi. **IQR:** The interquartile range is the difference between the 75th and 25th percentiles, showing how spread out the middle 50% of the data is. Makkah has the highest IQR, followed by Riyadh and the Eastern Region.
vii. **Skewness:** The skewness value shows the asymmetry of the data distribution. Riyadh has a positive skew, meaning a longer tail on the right. Makkah and the Eastern Region have relatively low skewness.
viii. **Kurtosis:** Makkah and the Eastern Region have negative kurtosis, meaning their distributions are flatter than a normal distribution, while Riyadh has a high kurtosis, indicating a more peaked distribution.

Next, we present a graphical representation of the data to provide a clearer visual understanding of the trends observed in the descriptive statistics. The following chart illustrates the number of tourists in Riyadh, Makkah, the Eastern Region, and the total of these three regions over the years.



As shown in the chart above, we can clearly observe the trends in tourist numbers across Riyadh, Makkah, and the Eastern Region over the years. The data reveals some important trends, including a significant decline in tourist numbers during 2020 and 2021, likely due to the COVID-19 pandemic. Additionally, Makkah consistently has higher tourist numbers compared to the other two regions, and the Eastern Region generally experiences more tourists than Riyadh.

3.2 Analysis of Tourist Numbers Using Time Series

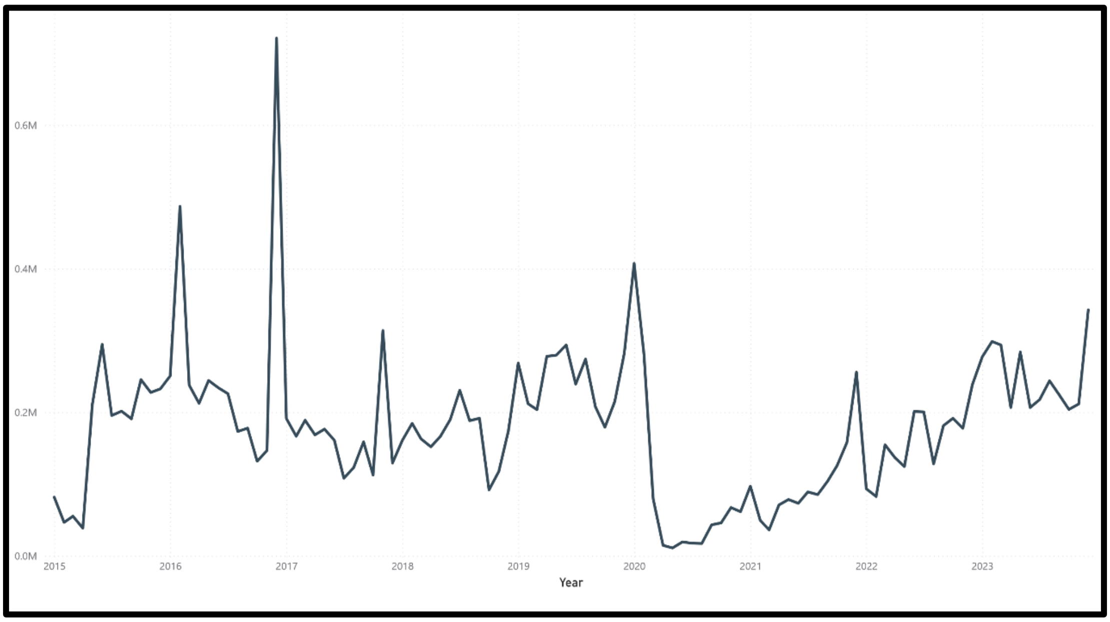
In this section, we will apply time series analysis to forecast the future trends in tourist numbers across Riyadh, Makkah, and the Eastern Region. By examining the historical data, we aim to understand the patterns and seasonal variations in tourist arrivals, while also predicting potential future trends for 2024 and 2025. This analysis will help to identify any recurring seasonal impacts, such as the effects of the Hajj season and Ramadan, and will provide valuable insights for decision-making in the tourism sector.

3.2.1 Forecast Analysis of Tourist Numbers in Riyadh

In this section, the time series data for the number of tourists visiting Riyadh from 2015 to 2023 is analyzed, with forecasts extending to 2025.

1. Time Series Plot for Riyadh (2015 - 2023)

The figure below illustrates the time series for the number of tourists in Riyadh, where the period from 2015 to 2023 represents the actual data.



2. Forecasting Model for 2024

To forecast the number of tourists in Riyadh for 2024, a SARIMA model was constructed. The model development process included the following steps:

2.1 Stationarity Testing

Before selecting the model, the time series was tested for stationarity using three tests:

• Augmented Dickey-Fuller (ADF) test: The p-value was **0.2777** , indicating non-stationary .

• Kwiatkowski-Phillips-Schmidt-Shin (KPSS) test: The p-value was **0.1**, indicating stationary .

• ARCH Test: The p-value was **0.793**, indicates that there is no significant evidence of heteroskedasticity.

These results suggest the need for data transformation or differencing to achieve stationarity before proceeding with model selection.

2.2 Data Transformation for Stationarity

The stationarity tests initially indicated that the time series was non-stationary. To resolve this, two regular differences were applied to eliminate the trend component. Following this transformation, the series became stationary, as confirmed by the stationarity test results.

The p-values of the stationarity tests after transformation are as follows:

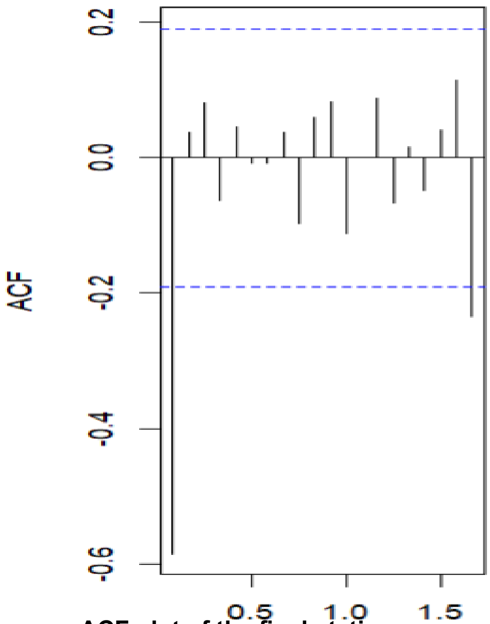
• Augmented Dickey-Fuller (ADF) test: **0.01** , indicating stationary .

• Kwiatkowski-Phillips-Schmidt-Shin (KPSS) test: **0.1 ,** indicating stationary.

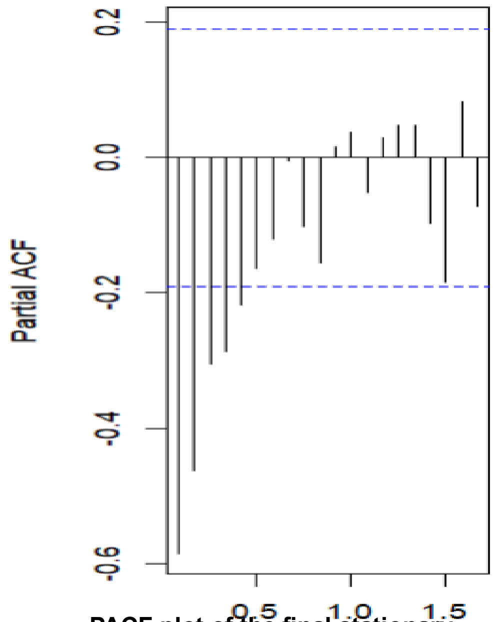
• ARCH test: **0.05987**, indicates that there is no significant evidence of heteroskedasticity.

2.3 Identification of Model Parameters Using ACF and PACF

The following ACF and PACF plots were generated after applying the necessary transformations to ensure stationarity in both mean and variance.



ACF plot of the final stationary series.



PACF plot of the final stationary series.

Suggested Value	Interpretation	Observation from plot	indicator
q=1	Suggests a Moving Average (MA) component of order 1	Cuts off after lag 1	ACF
p=0	The PACF gradually decays toward zero without a sharp cutoff, suggesting the absence of a clear AR structure.	gradually decays toward zero	PACF

2.4 Mathematical Representation of the Selected Model

The selected model is a seasonal ARIMA model (SARIMA) with the configuration **SARIMA(2,2,1)(0,1,1)[12].**

The mathematical formulation of the model is as follows:

$$\Delta^2(\Delta_{12} y_t) = -0.5029 \cdot \Delta^2(\Delta_{12} y_{t-1}) - 0.2948 \cdot \Delta^2(\Delta_{12} y_{t-2}) - 0.9974 \cdot \varepsilon_{t-1} - 0.9912 \cdot \varepsilon_{t-12} + \varepsilon_t$$

2.5 Coefficient Significance

This section presents the significance of the estimated coefficients based on the coeftest function. The p-values indicate whether each parameter is statistically different from zero.

Coefficient	Estimated	P-value
ar1	-0.502922	3.029e-07 ***
ar2	-0.294847	0.002531 **
ma1	-0.997396	< 2.2e-16 ***
sma1	-0.991184	6.407e-14 ***

All coefficients are statistically significant at the 5% level, which supports the validity of their inclusion in the model.

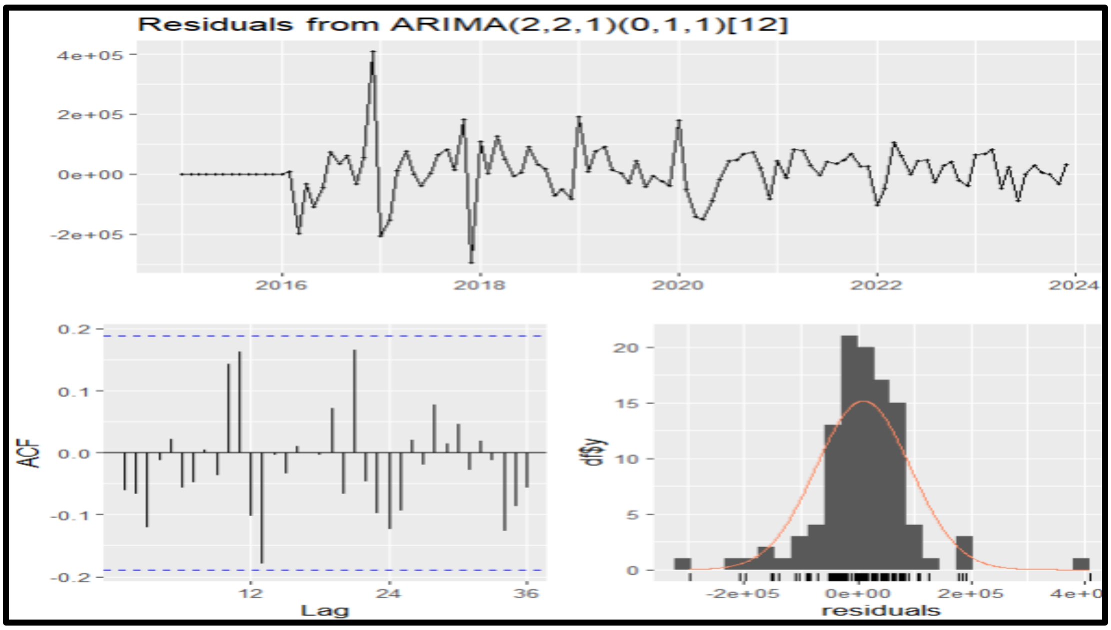
2.6 Model Diagnostics

After fitting the model, residuals were tested to validate model adequacy:

• Ljung-Box test: p-value = **0.7173** (indicating no autocorrelation).

• Runs test: p-value = **0.0737** (indicating randomness in the residuals).

• shapiro-test: p-value= **5.048e-07** (indicating the residuals does not follow normal)



Overall, the residual diagnostics indicated that the model was appropriate for forecasting.

2.7 Adjustment of Prediction Intervals to Avoid Negative Bounds

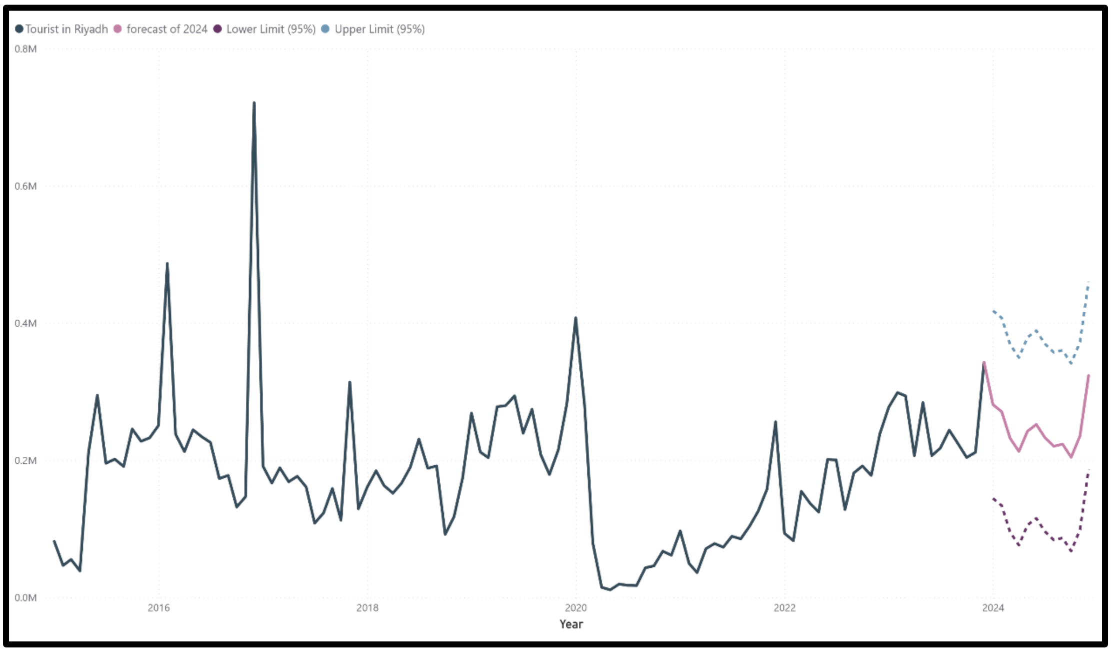
The original 95% prediction intervals generated by the SARIMA(2,2,1)(0,1,1)[12] model included some negative lower bounds, which are not meaningful in the context of tourism data, as the number of tourists cannot be negative. For instance, in certain forecasted months, the lower bounds dropped below zero, indicating a need for adjustment. To address this issue, custom 90% prediction intervals were calculated using the standard deviation of the residuals. This approach ensured all interval values

2.8 Forecast Results for 2024

The table below presents the forecasted values for each month in 2024, including the lower and upper confidence limits (90% interval):

Month	Forecast	Lower Limit (90%)	Upper Limit (90%)
January	307358	144283	417855
February	265324	133730	407302
March	221117	95290	368862
April	175242	75976	349548
May	213249	105431	379003
June	235519	115272	388844
July	217653	96148	369720
August	202437	83562	357134
September	236261	86579	360151
October	205747	67445	341017
November	258900	98661	372233
December	343495	186504	460076

The following chart visualizes the forecasted values for 2024, along with the corresponding confidence intervals. These intervals provide a range within which the actual values are expected to fall, offering a measure of uncertainty in the predictions. The forecast, based on historical data and the applied time series model, aims to provide insights into the expected tourist numbers in 2024, while the confidence limits help in understanding the potential variability and risk associated with the predictions.



2.9 Model Accuracy Evaluation

The forecasting performance of the selected SARIMA(2,2,1)(0,1,1)[12] model was evaluated using the Mean Absolute Percentage Error (MAPE). The model achieved a MAPE value of 8.79%, which reflects a high level of accuracy. Generally, a MAPE below 10% is considered to indicate a very good forecasting model, supporting the reliability of the model in capturing the underlying patterns of the data and providing dependable future estimates.

3. Forecasting Model for 2025

The forecast for 2025 is based on the time series model developed using historical data up until 2024. The forecast for 2024, which was treated as actual data due to the unavailability of real values for that year, serves as a proxy for the actual observations. By utilizing this approach, we can generate predictions for 2025, building on the most recent trends observed in the tourism data.

It is important to note that while the forecast for 2024 was treated as real data for the sake of analysis, it introduces an inherent limitation, as actual data for that year was not available at the time of this study. Despite this, the model provides valuable insights into the projected tourist numbers for 2025, offering a foundation for future planning and decision-making in the tourism sector.

3.1 Stationarity Testing

Before selecting the model, the time series was tested for stationarity using three tests:

• Augmented Dickey-Fuller (ADF) test: The p-value was **0.2573** , indicating non-stationary .

• Kwiatkowski-Phillips-Schmidt-Shin (KPSS) test: The p-value was **0.1**, indicating stationary .

• ARCH Test: The p-value was **0.6747**, indicates that there is no significant evidence of heteroskedasticity.

These results suggest the need for data transformation or differencing to achieve stationarity before proceeding with model selection.

3.2 Data Transformation for Stationarity

The stationarity tests initially indicated that the time series was non-stationary. To resolve this, two regular differences and one seasonal difference were applied to eliminate the trend component. Following this transformation, the series became stationary, as confirmed by the stationarity test results.

The p-values of the stationarity tests after transformation are as follows:

• Augmented Dickey-Fuller (ADF) test: **0.01** , indicating stationary .

• Kwiatkowski-Phillips-Schmidt-Shin (KPSS) test: **0.1 ,** indicating stationary.

• ARCH test: **3.417e-05 ,** indicates that there is no significant evidence of heteroskedasticity.

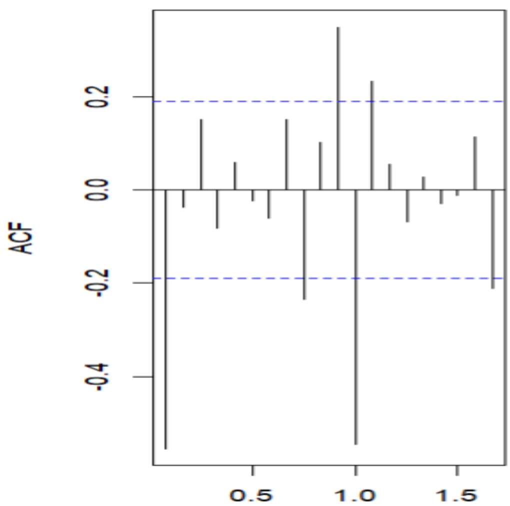
Although several transformations were applied to achieve stationarity — including two regular (non-seasonal) differences and one seasonal difference — the series still failed to pass formal stationarity tests. Thus, despite some visual indications from the ACF and PACF plots, the series could not be statistically confirmed as stationary.

As a next step, the model was estimated using the differenced series, and diagnostic checks were conducted on the residuals. The results indicated that the residuals were stationary, with no significant autocorrelation or conditional heteroscedasticity. This suggests that the model successfully captured the underlying structure of the data and transformed it into a stationary process through its specification.

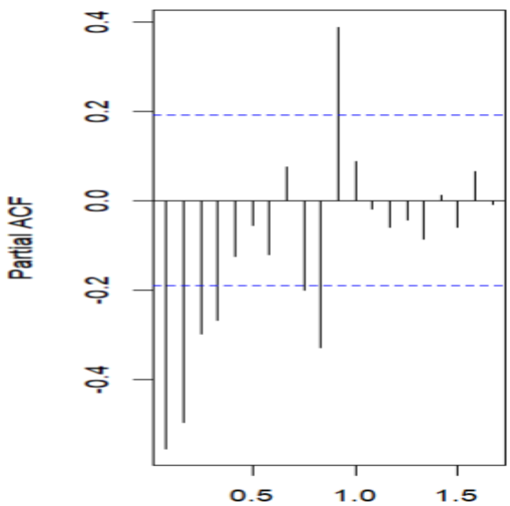
This approach is consistent with the recommendations found in time series literature. According to Shumway and Stoffer (2017), the adequacy of an ARIMA model is ultimately determined by the behavior of its residuals, which should resemble white noise. Similarly, Brockwell and Davis (2016) emphasize that even when the original series appears non-stationary, the model can still be valid if its residuals exhibit stationary white noise properties.

3.3 Identification of Model Parameters Using ACF and PACF

The following ACF and PACF plots were generated after applying the necessary transformations to ensure stationarity in both mean and variance.



ACF plot of the final stationary series.



PACF plot of the final stationary series.

Suggested Value	Interpretation	Observation from plot	indicator
q=1	Suggests a Moving Average (MA) component of order 1	Cuts off after lag 1	ACF
p=0	The PACF gradually decays toward zero without a sharp cutoff, suggesting the absence of a clear AR structure.	Cuts off after lag zero	PACF

3.4 Mathematical Representation of the Selected Model

The selected model is a seasonal ARIMA model (SARIMA) with the configuration **SARIMA(2,2,1)(1,1,1)[12]**.

The mathematical formulation of the model is as follows:

$$\Delta^2(\Delta_{12}y_t) = -0.5272\cdot\Delta^2(\Delta_{12}y_{t-1}) - 0.2747\cdot\Delta^2(\Delta_{12}y_{t-2}) - 0.9980\cdot\varepsilon_{t-1} - 0.2001\cdot\Delta^2(\Delta_{12}y_{t-12}) - 0.9867\cdot\varepsilon_{t-12} + \varepsilon_t$$

3.5 Coefficient Significance

This section presents the significance of the estimated coefficients based on the coeftest function. The p-values indicate whether each parameter is statistically different from zero.

Coefficient	Estimated	P-value
ar1	-0.527237	1.999e-08 ***
ar2	-0.274693	0.003188 **
ma1	-0.998032	< 2.2e-16 ***
sar1	-0.200107	0.045832 *
sma1	-0.986705	6.981e-10 ***

All coefficients are statistically significant at the 5% level, which supports the validity of their inclusion in the model.

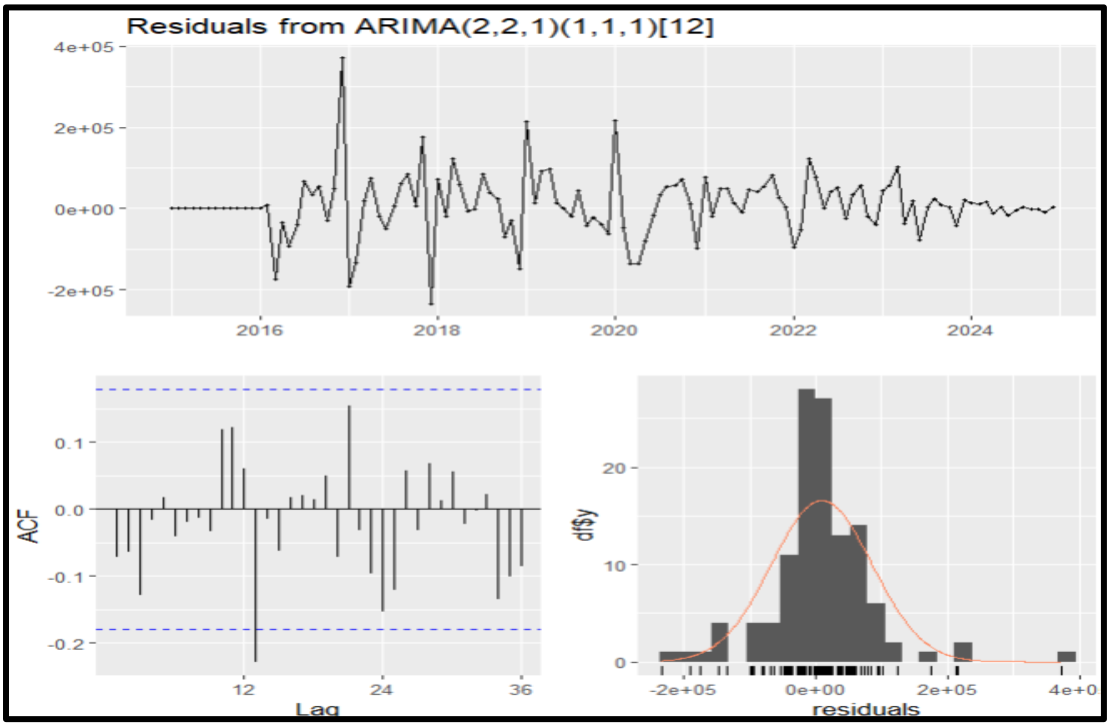
3.6 Model Diagnostics

After fitting the model, residuals were tested to validate model adequacy:

• Ljung-Box test: p-value = **0.1399** (indicating no autocorrelation).

• Runs test: p-value = **0.08168** (indicating randomness in the residuals).

• shapiro-test: p-value= **3.976e-07** (indicating the residuals does not follow normal)



• ACF and PACF plots showed no significant lags, except for **lag 13** which was tested using the Ljung-Box test. The p-values for this lag was **0.2951** , indicating no significant autocorrelation at this lag.

Overall, the residual diagnostics indicated that the model was appropriate for forecasting.

3.7 Adjustment of Prediction Intervals to Avoid Negative Bounds

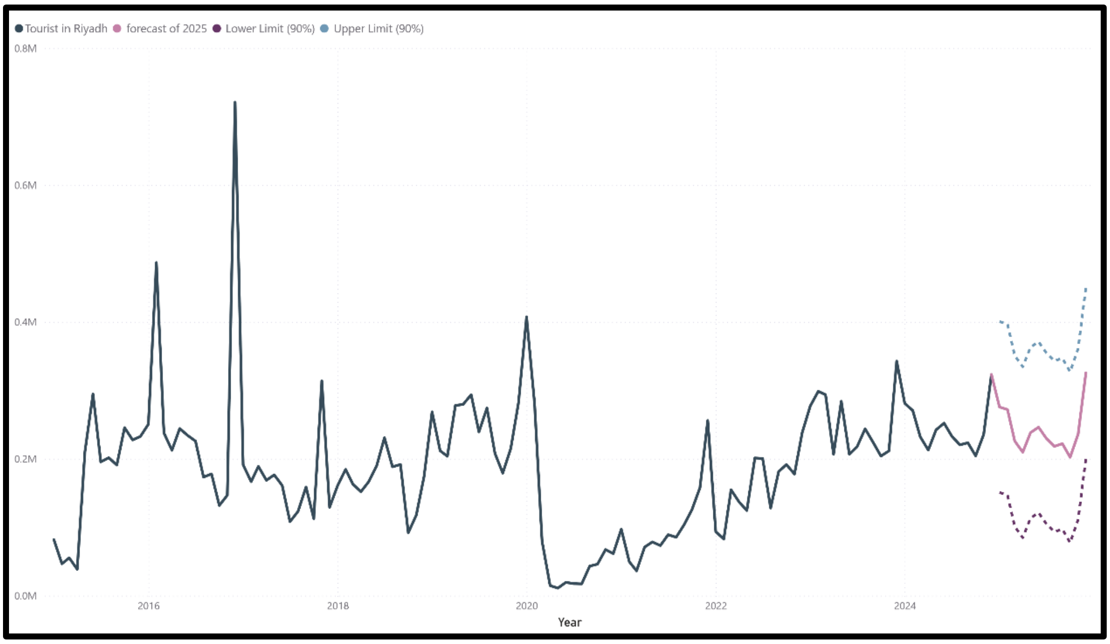
The original 95% prediction intervals generated by the SARIMA(2,2,1)(1,1,1)[12] model included some negative lower bounds, which are not meaningful in the context of tourism data, as the number of tourists cannot be negative. For instance, in certain forecasted months, the lower bounds dropped below zero, indicating a need for adjustment. To address this issue, custom 90% prediction intervals were calculated using the standard deviation of the residuals. This approach ensured all interval values

3.8 Forecast Results for 2025

The table below presents the forecasted values for each month in 2025, including the lower and upper confidence limits (90% interval):

Month	Forecast	Lower Limit (90%)	Upper Limit (90%)
January	275643.2	150737	400550
February	271928.3	147022	396835
March	225902.1	100995	350809
April	209583.1	84676	334490
May	238187.1	113280	363094
June	246145.1	121238	371052
July	229864.1	104957	354771
August	218169.9	93263	343077
September	222031.9	97125	346938
October	202183.3	77277	327090
November	236191.3	111285	361098
December	325685.4	200779	450592

The following chart visualizes the forecasted values for 2025, along with the corresponding confidence intervals. These intervals provide a range within which the actual values are expected to fall, offering a measure of uncertainty in the predictions. The forecast, based on historical data and the applied time series model, aims to provide insights into the expected tourist numbers in 2025, while the confidence limits help in understanding the potential variability and risk associated with the predictions.



3.9 Model Accuracy Evaluation

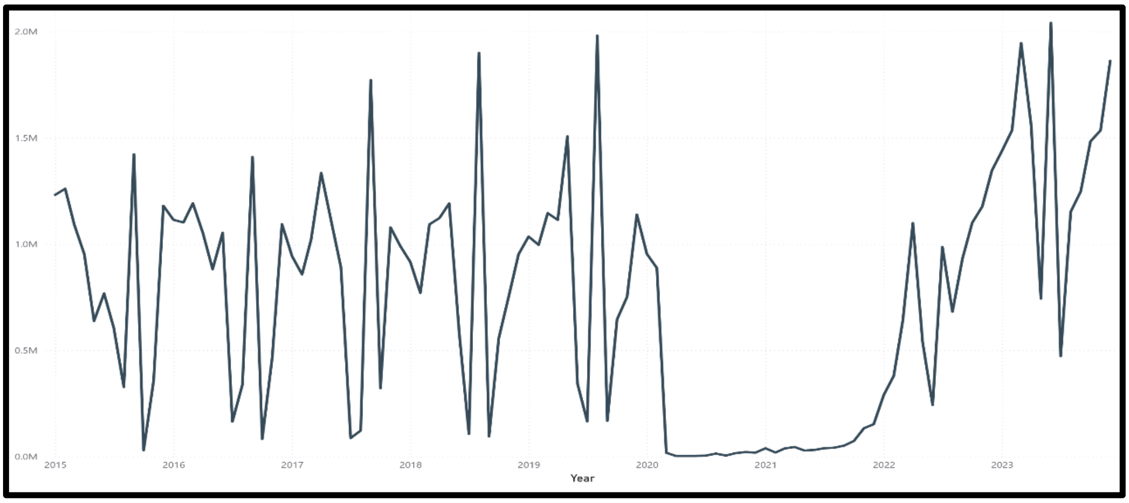
The forecasting performance of the selected SARIMA(2,2,1)(1,1,1)[12] model was evaluated using the Mean Absolute Percentage Error (MAPE). The model achieved a MAPE value of **8.64**%, which reflects a high level of accuracy. Generally, a MAPE below 10% is considered to indicate a very good forecasting model, supporting the reliability of the model in capturing the underlying patterns of the data and providing dependable future estimates.

3.2.2 FORECAST ANALYSIS OF TOURIST NUMBERS IN MAKKAH

In this section, the time series data for the number of tourists visiting Makkah from 2015 to 2023 is analyzed, with forecasts extending to 2025.

1. Time Series Plot for Makkah (2015 - 2023)

The figure below illustrates the time series for the number of tourists in Makkah, where the period from 2015 to 2023 represents the actual data.



2. Incorporating the Effects of the COVID-19 Pandemic on Tourist Arrivals in Makkah

To account for the impact of the COVID-19 pandemic on tourism in Makkah, a dummy variable was included in the model for the period from January 2020 to May 2022. This variable was also incorporated when forecasting for both 2024 and 2025. The rationale behind this inclusion is that tourism in Makkah was significantly more affected by the pandemic compared to other regions, primarily due to the sharp decline in religious tourism. Under normal circumstances, Makkah receives over two million visitors annually, especially during Hajj and Umrah seasons, which were severely disrupted during the pandemic.

3. Forecasting Model for 2024

To forecast the number of tourists in Makkah for 2024, a SARIMA model was constructed. The model development process included the following steps:

3.1 Stationarity Testing

Before selecting the model, the time series was tested for stationarity using three tests:

• Augmented Dickey-Fuller (ADF) test: The p-value was **0.7865**, indicating non-stationary .

• Kwiatkowski-Phillips-Schmidt-Shin (KPSS) test: The p-value was **0.1**, indicating stationary .

• ARCH Test: The p-value was **0.0003427** , indicating the presence of heteroscedasticity in the time series.

These results suggest the need for data transformation or differencing to achieve stationarity before proceeding with model selection.

3.2 Data Transformation for Stationarity

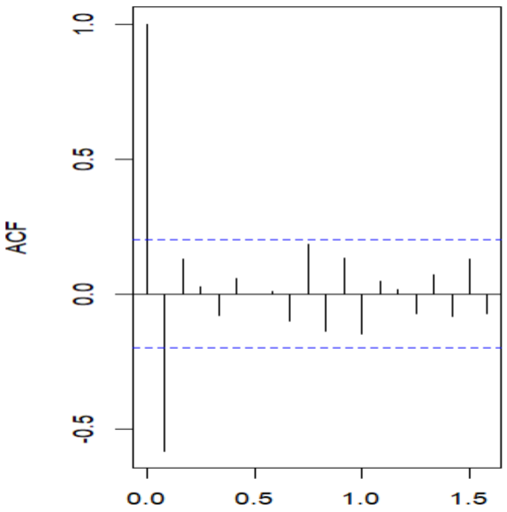
After conducting stationarity tests, the time series was found to be non-stationary. To address this issue, a Box-Cox transformation was first applied to stabilize the variance. Then, both seasonal and regular differencing were performed to remove trends and seasonal patterns. Following these transformations, the time series became stationary, as confirmed by the results of the stationarity tests.

The p-values of the stationarity tests after transformation are as follows:

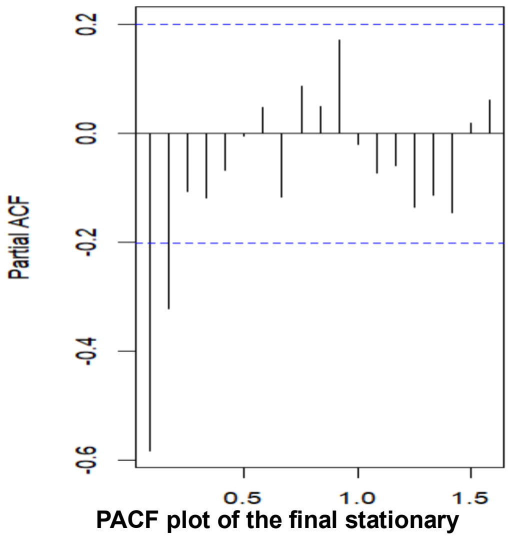
- Augmented Dickey-Fuller (ADF) test: **0.01** , indicating stationary .
- Kwiatkowski-Phillips-Schmidt-Shin (KPSS) test: **0.1 ,** indicating stationary.
- ARCH test: **0.0515** ,indicates that there is no significant evidence of heteroskedasticity.

3.3 Identification of Model Parameters Using ACF and PACF

The following ACF and PACF plots were generated after applying the necessary transformations to ensure stationarity in both mean and variance.



ACF plot of the final stationary series.



PACF plot of the final stationary series.

Suggested Value	Interpretation	Observation from plot	indicator
q=2	Suggests a Moving Average (MA) component of order 2	Cuts off after lag 2	ACF
p=2	Suggests an AutoRegressive (AR) component of order 2	Cuts off after lag 2	PACF

3.4 Mathematical Representation of the Selected Model

The selected model is a seasonal ARIMA model (SARIMA) with the configuration **SARIMA(2,1,3)(0,1,0)[12].**

The mathematical formulation of the model is as follows:

$$\Delta yt = -0.4818\Delta yt-1-0.8207\Delta yt-2-0.4783\varepsilon t-1+0.5433\varepsilon t-2-0.9560\varepsilon t-3+\varepsilon t$$

3.5 Coefficient Significance

This section presents the significance of the estimated coefficients based on the coeftest function. The p-values indicate whether each parameter is statistically different from zero.

Coefficient	Estimated	P-value
ar1	-4.8178e-01	2.459e-08 ***
ar2	-8.2075e-01	< 2.2e-16 ***
ma1	-4.7826e-01	1.394e-13 ***
ma2	5.4330e-01	< 2.2e-16 ***
ma3	-9.5598e-01	< 2.2e-16 ***
xreg	-2.4219e+04	1.225e-12 ***

All coefficients are statistically significant at the 5% level, which supports the validity of their inclusion in the model.

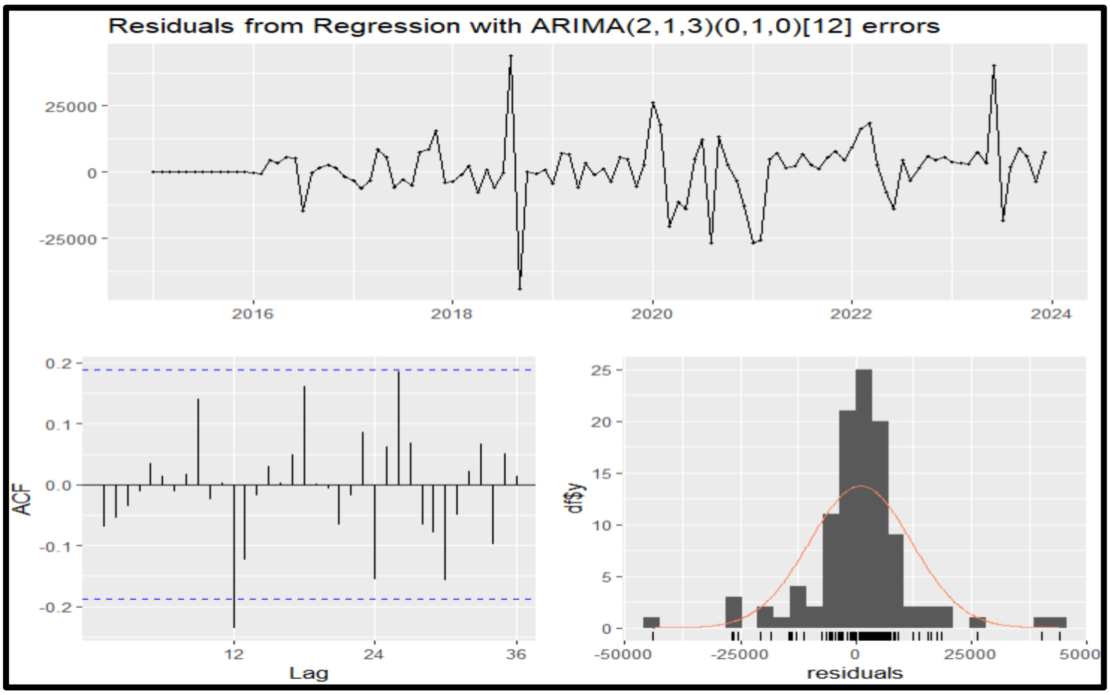
3.6 Model Diagnostics

After fitting the model, residuals were tested to validate model adequacy:

- Ljung-Box test: p-value = **0.453** (indicating no autocorrelation).

• Runs test: p-value = **0.09085** (indicating randomness in the residuals).

• shapiro-test: p-value= **2.332e-08** (indicating the residuals does not follow normal)



• ACF and PACF plots showed no significant lags, except for **lag 12**, and **lag 26**, which were tested using the Ljung-Box test. The p-values for these lags were **0.5687** , and **0.4087** respectively, indicating no significant autocorrelation at these lags.

Overall, the residual diagnostics indicated that the model was appropriate for forecasting.

3.7 Adjustment of Prediction Intervals to Avoid Negative Bounds

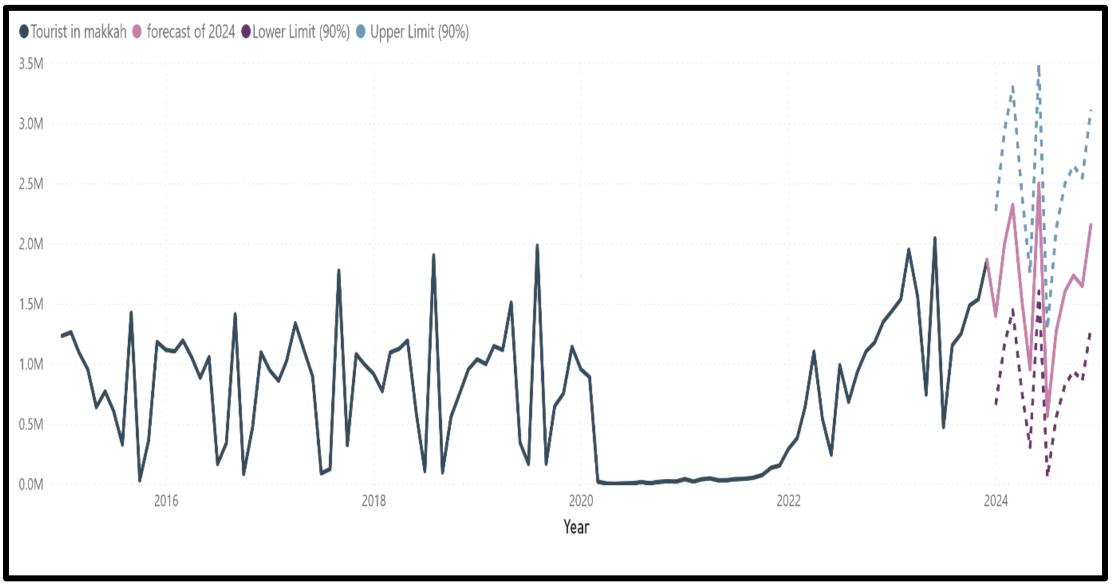
The original 95% prediction intervals generated by the SARIMA(2,1,3)(0,1,0)[12] model included some negative lower bounds, which are not meaningful in the context of tourism data, as the number of tourists cannot be negative. For instance, in certain forecasted months, the lower bounds dropped below zero, indicating a need for adjustment. To address this issue, custom 90% prediction intervals were calculated using the standard deviation of the residuals. This approach ensured all interval values

3.8 Forecast Results for 2024

The table below presents the forecasted values for each month in 2024, including the lower and upper confidence limits (90% interval):

Month	Forecast	Lower Limit (90%)	Upper Limit (90%)
January	1397775	656414	2267878
February	1996196	1160995	2939828
March	2319929	1443537	3297276
April	1550674	782466	2441292
May	950949	306789	1751076
June	2496682	1599762	3491111
July	564271	52330	1283904
August	1270046	553173	2121849
September	1600281	823856	2497267
October	1731884	934674	2645152
November	1643834	860373	2546303
December	2150685	1295188	3110833

The following chart visualizes the forecasted values for 2024, along with the corresponding confidence intervals. These intervals provide a range within which the actual values are expected to fall, offering a measure of uncertainty in the predictions. The forecast, based on historical data and the applied time series model, aims to provide insights into the expected tourist numbers in 2024, while the confidence limits help in understanding the potential variability and risk associated with the predictions.



3.9 Model Accuracy Evaluation

The forecasting performance of the selected SARIMA(2,1,3)(0,1,0)[12] model was evaluated using the Mean Absolute Percentage Error (MAPE). The model achieved a MAPE value of 16.69%, which reflects a high level of accuracy. Generally, a MAPE below 10% is considered excellent, while values between 10% and 20% are still acceptable and indicate a reasonably good forecasting model. In this case, the MAPE of 16.69% suggests that the model captures the underlying patterns of the data with a fair level of accuracy and provides reasonably reliable future estimates.

4. Forecasting Model for 2025

The forecast for 2025 is based on the time series model developed using historical data up until 2024. The forecast for 2024, which was treated as actual data due to the unavailability of real values for that year, serves as a proxy for the actual observations. By utilizing this approach, we can generate predictions for 2025, building on the most recent trends observed in the tourism data.

It is important to note that while the forecast for 2024 was treated as real data for the sake of analysis, it introduces an inherent limitation, as actual data for that year was not available at the time of this study. Despite this, the model provides valuable insights into the projected tourist numbers for 2025, offering a foundation for future planning and decision-making in the tourism sector.

4.1 Stationarity Testing

Before selecting the model, the time series was tested for stationarity using three tests:

• Augmented Dickey-Fuller (ADF) test: The p-value was **0.6829** , indicating non-stationary .

• Kwiatkowski-Phillips-Schmidt-Shin (KPSS) test: The p-value was **0.04603**, indicating non-stationary .

• ARCH Test: The p-value was **1.262e-08** , indicating the presence of heteroscedasticity in the time series.

These results suggest the need for data transformation or differencing to achieve stationarity before proceeding with model selection.

4.2 Data Transformation for Stationarity

After conducting stationarity tests, the time series was found to be non-stationary. To address this issue, two regular differences and two seasonal differences were applied to eliminate both trend and seasonal patterns. Following these transformations, the time series became stationary, as confirmed by the results of the stationarity tests.

The p-values of the stationarity tests after transformation are as follows:

• Augmented Dickey-Fuller (ADF) test: **0.01** , indicating stationary .

• Kwiatkowski-Phillips-Schmidt-Shin (KPSS) test: **0.1 ,** indicating stationary.

• ARCH test: **1.672e-06** , indicating the presence of heteroscedasticity in the time series.

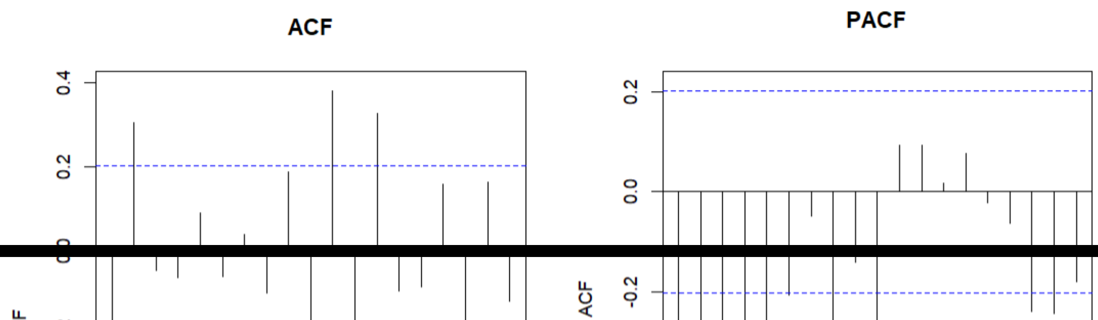
After fitting a time series model that includes relevant explanatory variables—such as a dummy variable accounting for the COVID-19 period—it is essential to examine the residuals to assess whether the model adequately captures the structure of the data. Specifically, to evaluate the presence of conditional heteroskedasticity, the ARCH test should be applied to the residuals, not the original data.

This approach is consistent with standard econometric practice, where residual diagnostics are used to verify model adequacy. As noted by Enders (2015), "If the residuals from an estimated model exhibit conditional heteroskedasticity (as detected by the ARCH test), this implies that a model such as GARCH may be more appropriate." Therefore, if the ARCH test on the residuals yields a non-significant result (p-value > 0.05), it indicates that the model—including the dummy variable—successfully explains the time-varying variance, and the variance can be considered stable after modeling.

In summary, the stability of the variance after including the dummy variable is validated not by testing the raw data, but by testing the residuals of the fitted model.

4.3 Identification of Model Parameters Using ACF and PACF

The following ACF and PACF plots were generated after applying the necessary transformations to ensure stationarity in both mean and variance.



ACF plot of the final stationary series.

PACF plot of the final stationary series.

Suggested Value	**Interpretation**	**Observation from plot**	**indicator**
q=2	Suggests a Moving Average (MA) component of order 2	Cuts off after lag 2	ACF
p=0	The PACF gradually decays toward zero without a sharp cutoff, suggesting the absence of a clear AR structure.	gradually decays toward zero	PACF

The ACF and PACF plots show significant spikes at specific lags, indicating the presence of seasonality in the time series data.

4.4 Mathematical Representation of the Selected Model

The selected model is a seasonal ARIMA model (SARIMA) with the configuration **SARIMA(0,2,2)(2,2,0)[12].**

The mathematical formulation of the model is as follows:The mathematical formulation of the model is as follows:

$$\Delta^2 y_t = -1.8550\varepsilon_{t-1} + 0.8760\varepsilon_{t-2} - 0.7023\varepsilon_{t-12} - 0.4970\varepsilon_{t-24} + \varepsilon_t - 669465.0\cdot x_t$$

3.5 Coefficient Significance

Coefficient	Estimated	P-value
ma1	-1.8550e+00	< 2.2e-16 ***

ma2	8.7598e-01	< 2.2e-16 ***
sar1	-7.0229e-01	1.576e-13 ***
sar2	-4.9696e-01	1.125e-07 ***
xreg	-6.6947e+05	2.409e-06 ***

This section presents the significance of the estimated coefficients based on the coeftest function. The p-values indicate whether each parameter is statistically different from zero.

All coefficients are statistically significant at the 5% level, which supports the validity of their inclusion in the model.

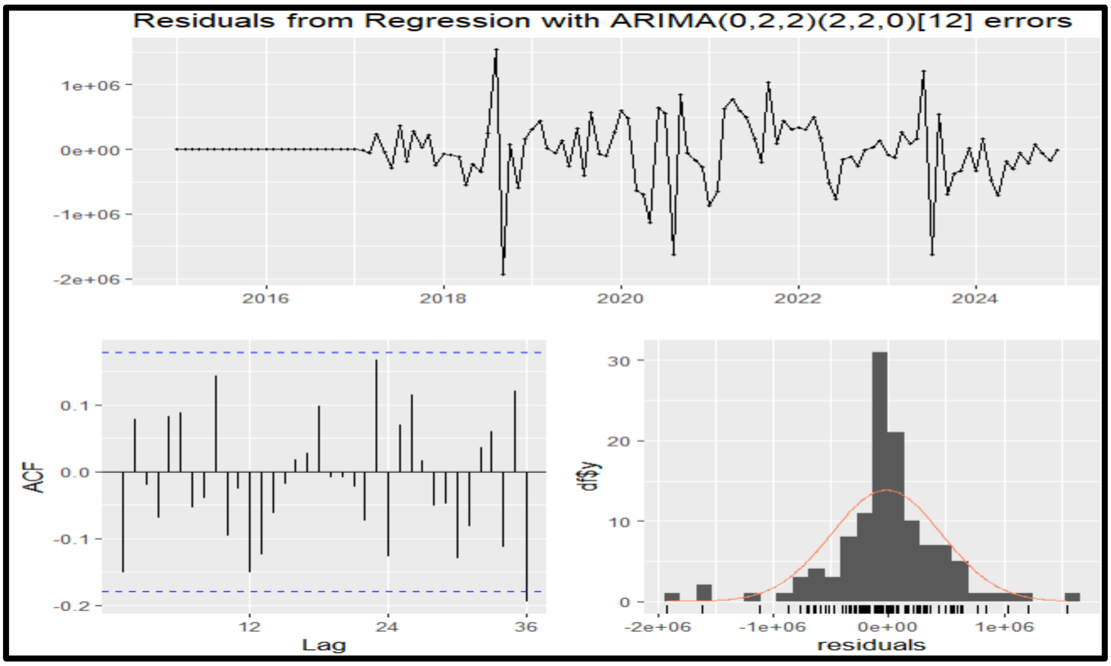
4.6 Model Diagnostics

After fitting the model, residuals were tested to validate model adequacy:

• Ljung-Box test: p-value = **0.1734** (indicating no autocorrelation).

• Runs test: p-value = **0.1357** (indicating randomness in the residuals).

• shapiro-test: p-value= **4.801e-07** (indicating the residuals does not follow normal)



• ACF and PACF plots showed no significant lags, except for **lag 36**, which were tested using the Ljung-Box test. The p-values for these lags were **0.1357** , indicating no significant autocorrelation at this lag.

Overall, the residual diagnostics indicated that the model was appropriate for forecasting.

4.7 Adjustment of Prediction Intervals to Avoid Negative Bounds

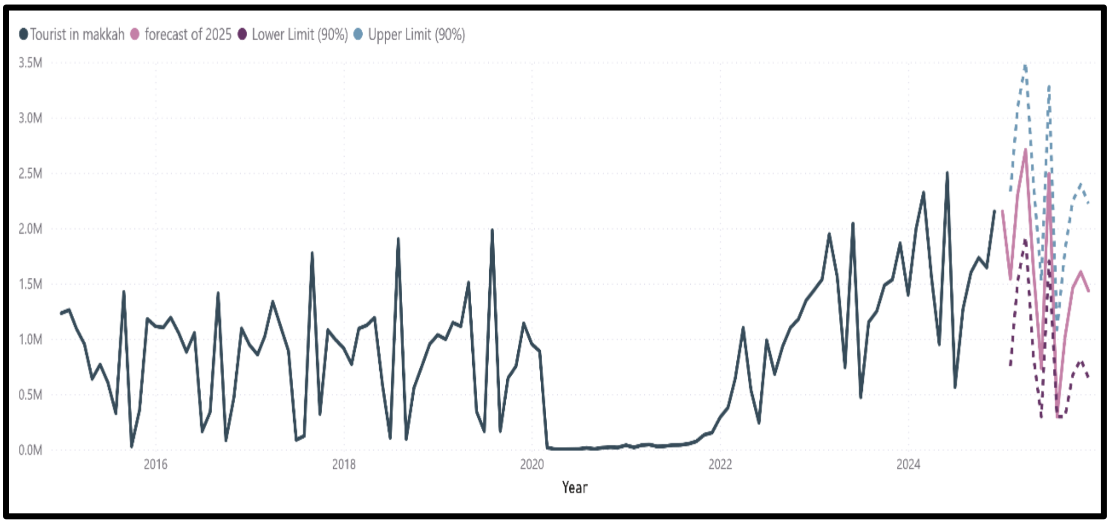
The original 95% prediction intervals generated by the SARIMA(0,2,2)(2,2,0)[12] model included some negative lower bounds, which are not meaningful in the context of tourism data, as the number of tourists cannot be negative. For instance, in certain forecasted months, the lower bounds dropped below zero, indicating a need for adjustment. To address this issue, custom 90% prediction intervals were calculated using the standard deviation of the residuals. This approach ensured all interval values

4.8 Forecast Results for 2025

The table below presents the forecasted values for each month in 2025, including the lower and upper confidence limits (90% interval):

Month	Forecast	Lower Limit (90%)	Upper Limit (90%)
January	1541758.4	754140	2329376
February	2302299.9	1514682	3089918
March	2707689.7	1920072	3495308
April	1636565.6	848948	2424184
May	734965	296543	1522583
June	2489817.8	1702200	3277436
July	296542.6	296543	1084161
August	1015986.5	296543	1803604
September	1460233.5	672616	2247851
October	1604049	816431	2391667
November	1434178.3	646560	2221796
December	2058419.1	1270801	2846037

The following chart visualizes the forecasted values for 2025, along with the corresponding confidence intervals. These intervals provide a range within which the actual values are expected to fall, offering a measure of uncertainty in the predictions. The forecast, based on historical data and the applied time series model, aims to provide insights into the expected tourist numbers in 2025, while the confidence limits help in understanding the potential variability and risk associated with the predictions.



4.9 Model Accuracy Evaluation

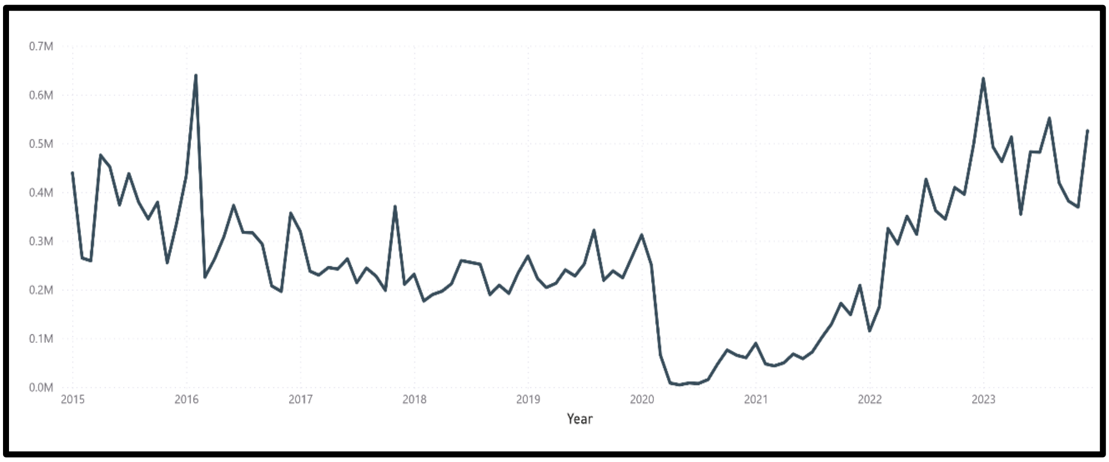
The forecasting performance of the selected SARIMA(0,2,2)(2,2,0)[12] model was evaluated using the Mean Absolute Percentage Error (MAPE). The model achieved a MAPE value of 17.96%, which reflects a high level of accuracy. Generally, a MAPE below 10% is considered excellent, while values between 10% and 20% are still acceptable and indicate a reasonably good forecasting model. In this case, the MAPE of 17.96% suggests that the model captures the underlying patterns of the data with a fair level of accuracy and provides reasonably reliable future estimates.

3.2.3 FORECAST ANALYSIS OF TOURIST NUMBERS IN EASTERN REGION

In this section, the time series data for the number of tourists visiting Eastern Region from 2015 to 2023 is analyzed, with forecasts extending to 2025.

1. Time Series Plot for Eastern Region (2015 - 2023)

The figure below illustrates the time series for the number of tourists in Eastern Region, where the period from 2015 to 2023 represents the actual data.



2. Forecasting Model for 2024

To forecast the number of tourists in Eastern Region for 2024, a SARIMA model was constructed. The model development process included the following steps:

2.1 Stationarity Testing

Before selecting the model, the time series was tested for stationarity using three tests:

• Augmented Dickey-Fuller (ADF) test: The p-value was **0.9157**, indicating non-stationary .

• Kwiatkowski-Phillips-Schmidt-Shin (KPSS) test: The p-value was **0.08218**, indicating stationary .

• ARCH Test: The p-value was **1.416e-08**, indicating the presence of heteroscedasticity in the time series.

These results suggest the need for data transformation or differencing to achieve stationarity before proceeding with model selection.

2.2 Data Transformation for Stationarity

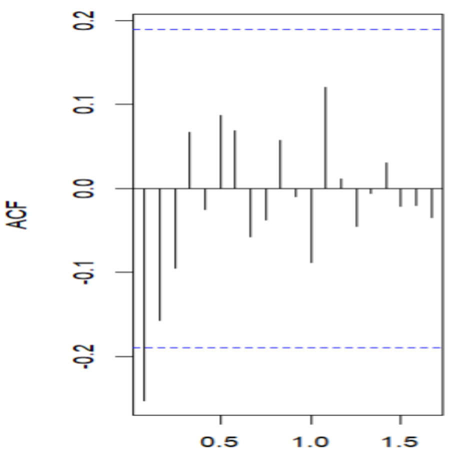
After conducting stationarity tests, the time series was found to be non-stationary. To address this, a first-order regular differencing was applied to the data. This transformation helped to remove the trend and stabilize the variance. Following this differencing, the time series became stationary, as confirmed by the results of the stationarity tests.

The p-values of the stationarity tests after transformation are as follows:

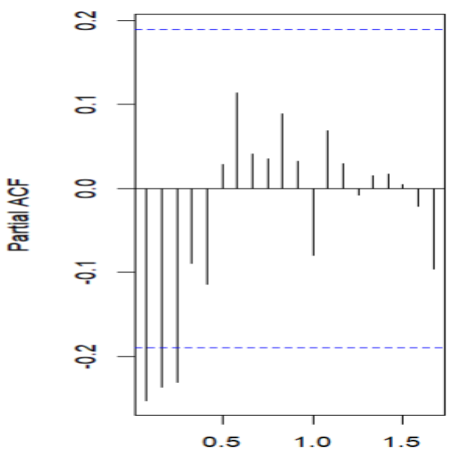
- Augmented Dickey-Fuller (ADF) test: **0.01** , indicating stationary .
- Kwiatkowski-Phillips-Schmidt-Shin (KPSS) test: **0.1 ,** indicating stationary.
- ARCH test: **0.4432** ,indicates that there is no significant evidence of heteroskedasticity.

2.3 Identification of Model Parameters Using ACF and PACF

The following ACF and PACF plots were generated after applying the necessary transformations to ensure stationarity in both mean and variance.



ACF plot of the final stationary series.



PACF plot of the final stationary series.

Suggested Value	Interpretation	Observation from plot	indicator
q=2	Suggests a Moving Average (MA) component of order 2	Cuts off after lag 2	ACF
p=3	Suggests an AutoRegressive (AR) component of order 3	Cuts off after lag 3	PACF

2.4 Mathematical Representation of the Selected Model

The selected model is a seasonal ARIMA model (SARIMA) with the configuration **SARIMA(2,1,2)(1,2,0)[12].**

The mathematical formulation of the model is as follows:

$$\Delta yt=1.2953\Delta yt-1-0.6404\Delta yt-2-1.7120\epsilon t-1+0.9684\epsilon t-2$$
$$-0.6355\Delta 2yt-12+\epsilon t$$

2.5 Coefficient Significance

This section presents the significance of the estimated coefficients based on the coeftest function. The p-values indicate whether each parameter is statistically different from zero.

Coefficient	Estimated	P-value
ar1	1.29531	< 2.2e-16 ***
ar2	-0.64037	4.637e-08 ***
ma1	-1.71205	< 2.2e-16 ***
ma2	0.96836	3.501e-16 ***
sar1	-0.63547	1.234e-08 ***

All coefficients are statistically significant at the 5% level, which supports the validity of their inclusion in the model.

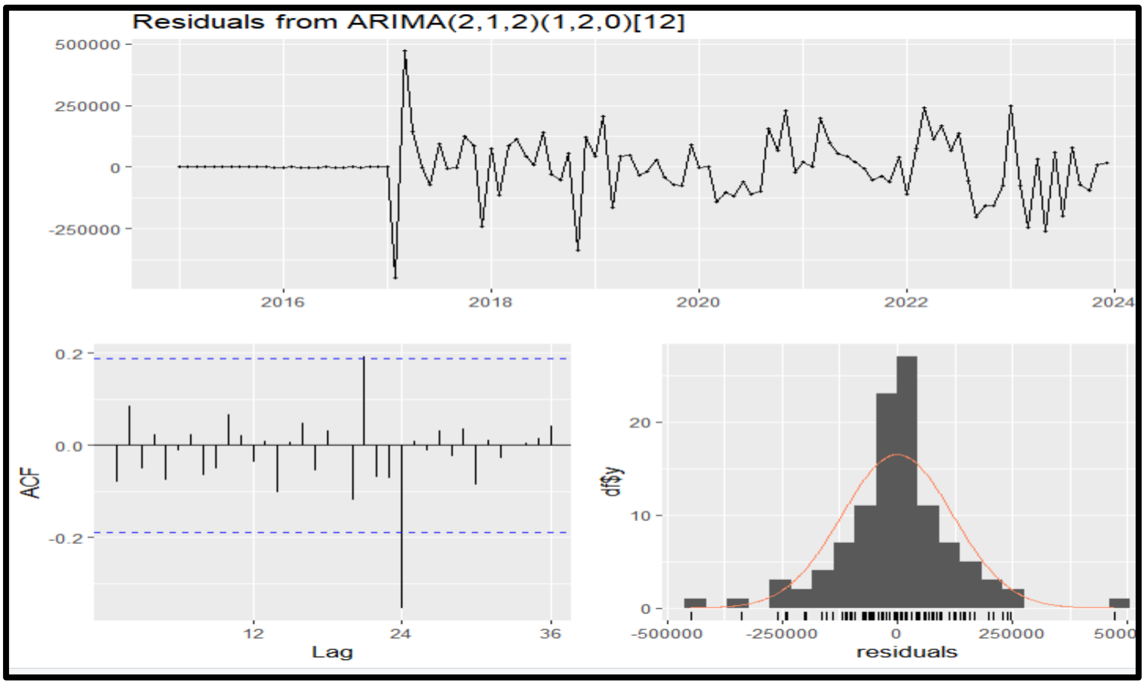
2.5 Model Diagnostics

After fitting the model, residuals were tested to validate model adequacy:

- Ljung-Box test: p-value = **0.6671** (indicating no autocorrelation).

• Runs test: p-value = **0.1249** (indicating randomness in the residuals).

• shapiro-test: p-value= **3.352e-05** (indicating the residuals does not follow normal)



• ACF and PACF plots showed no significant lags, except for **lag 21**, and **lag 24**, which were tested using the Ljung-Box test. The p-values for these lags were **0.8966**, and **0.1196** respectively, indicating no significant autocorrelation at these lags.

Overall, the residual diagnostics indicated that the model was appropriate for forecasting.

2.6 Adjustment of Prediction Intervals to Avoid Negative Bounds

The original 95% prediction intervals generated by the SARIMA(2,1,2)(1,2,0)[12] model included some negative lower bounds, which are not meaningful in the context of tourism data, as the number of tourists cannot be negative. For instance, in certain forecasted months, the lower bounds dropped below zero, indicating a need for adjustment. To address this issue, custom 90% prediction intervals were calculated using the standard deviation of the residuals. This approach ensured all interval values remained positive while maintaining the integrity and interpretability of the forecasts.

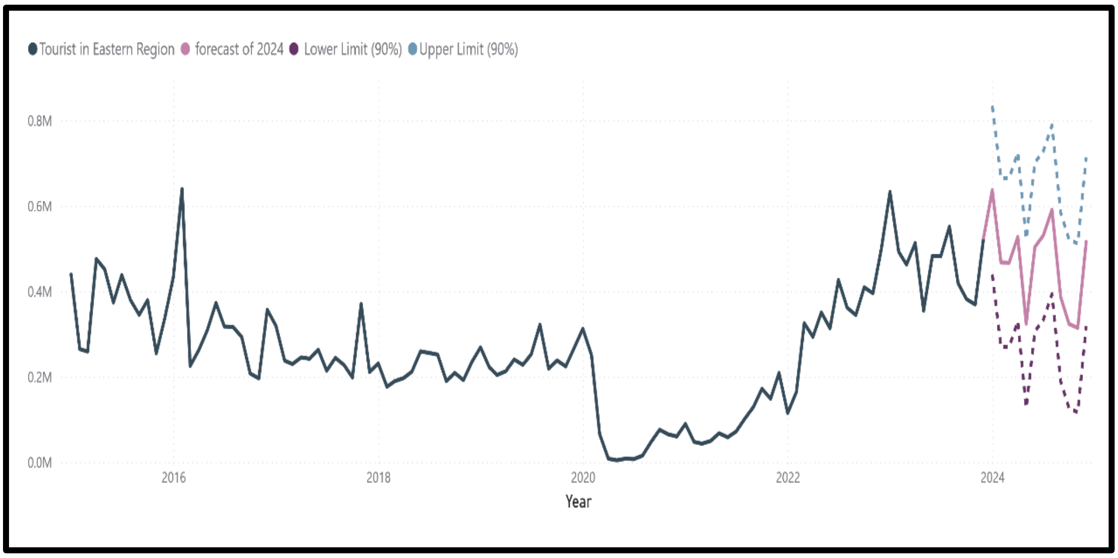
2.7 Forecast Results for 2024

The table below presents the forecasted values for each month in 2024, including the lower and upper confidence limits (90% interval):

Month	Forecast	Lower Limit (90%)	Upper Limit (90%)

January	636920.6	439397	834445
February	467501	269977	665025
March	466889.8	269366	664414
April	527369.5	329845	724894
May	324241.8	126718	521766
June	503815	306291	701339
July	530714	333190	728238
August	591036	393512	788560
September	386571.6	189048	584096
October	323395.7	125872	520920
November	314412	116888	511936
December	515863.7	318340	713388

The following chart visualizes the forecasted values for 2024, along with the corresponding confidence intervals. These intervals provide a range within which the actual values are expected to fall, offering a measure of uncertainty in the predictions. The forecast, based on historical data and the applied time series model, aims to provide insights into the expected tourist numbers in 2024, while the confidence limits help in understanding the potential variability and risk associated with the predictions.



2.8 Model Accuracy Evaluation

The forecasting performance of the selected SARIMA(2,1,2)(1,2,0)[12] model was evaluated using the Mean Absolute Percentage Error (MAPE). The model achieved a MAPE value of 6.62%, which reflects a high level of accuracy. Generally, a MAPE below 10% is considered to indicate a very good forecasting model, supporting the reliability of the model in capturing the underlying patterns of the data and providing dependable future estimates.

3. Forecasting Model for 2025

The forecast for 2025 is based on the time series model developed using historical data up until 2024. The forecast for 2024, which was treated as actual data due to the unavailability of real values for that year, serves as a proxy for the actual observations. By utilizing this approach, we can generate predictions for 2025, building on the most recent trends observed in the tourism data.

It is important to note that while the forecast for 2024 was treated as real data for the sake of analysis, it introduces an inherent limitation, as actual data for that year was not available at the time of this study. Despite this, the model provides valuable insights into the projected tourist numbers for 2025, offering a foundation for future planning and decision-making in the tourism sector.

3.1 Stationarity Testing

Before selecting the model, the time series was tested for stationarity using three tests:

• Augmented Dickey-Fuller (ADF) test: The p-value was **0.7609**, indicating non-stationary .

• Kwiatkowski-Phillips-Schmidt-Shin (KPSS) test: The p-value was **0.02877** , indicating stationary .

• ARCH Test: The p-value was **2.272e-10** , indicating the presence of heteroscedasticity in the time series.

These results suggest the need for data transformation or differencing to achieve stationarity before proceeding with model selection.

3.2 Data Transformation for Stationarity

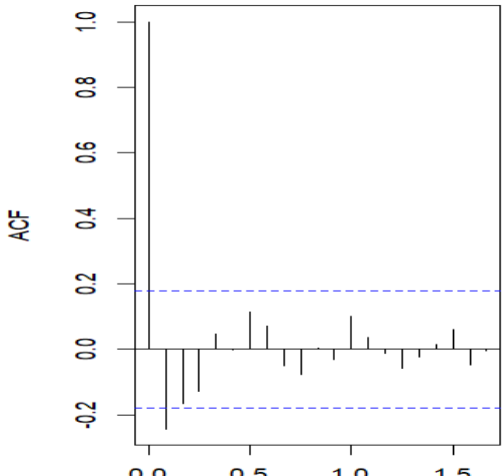
After conducting stationarity tests, the time series was found to be non-stationary. To address this, differencing was applied to the data. The differencing helped to remove trends and stabilized the variance. Following these transformation, the time series became stationary, as confirmed by the results of the stationarity tests.

The p-values of the stationarity tests after transformation are as follows:

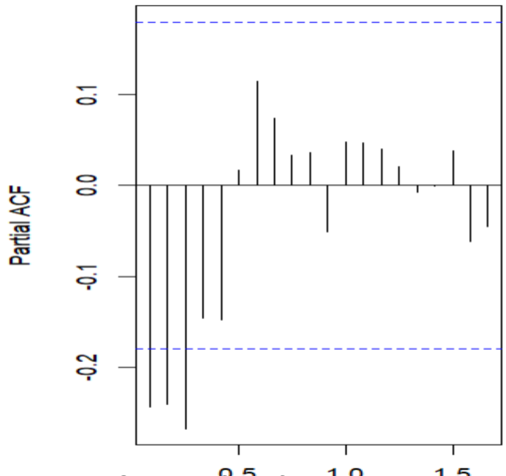
- Augmented Dickey-Fuller (ADF) test: **0.01** , indicating stationary .
- Kwiatkowski-Phillips-Schmidt-Shin (KPSS) test: **0.1 ,** indicating stationary.
- ARCH test: **0.3515** ,indicates that there is no significant evidence of heteroskedasticity.

3.3 Identification of Model Parameters Using ACF and PACF

The following ACF and PACF plots were generated after applying the necessary transformations (e.g., first differencing) to ensure stationarity in both mean and variance.



ACF plot of the final stationary series.



PACF plot of the final stationary series.

Suggested Value	Interpretation	Observation from plot	indicator
q=2	Suggests a Moving Average (MA) component of order 2	Cuts off after lag 2	ACF
p=3	Suggests an AutoRegressive (AR) component of order 3	Cuts off after lag 3	PACF

3.4 Mathematical Representation of the Selected Model

The selected model is a seasonal ARIMA model (SARIMA) with the configuration **SARIMA(2,1,2)(0,2,2)[12].**

The mathematical formulation of the model is as follows:

$$\Delta yt = 1.2555\Delta yt-1 - 0.3516\Delta yt-2 - 1.7004\epsilon t-1 + 0.7794\epsilon t-2 - 1.5096\Delta 2yt-12 + 0.7479\epsilon t$$

3.5 Coefficient Significance

This section presents the significance of the estimated coefficients based on the coeftest function. The p-values indicate whether each parameter is statistically different from zero.

Coefficient	Estimated	P-value
ar1	1.25552	3.034e-09 ***
ar2	-0.35160	0.02907 *
ma1	-1.70041	< 2.2e-16 ***
ma2	0.77939	2.104e-10 ***
sma1	-1.50962	1.900e-09 ***
sma2	0.74785	0.01369 *

All coefficients are statistically significant at the 5% level, which supports the validity of their inclusion in the model.

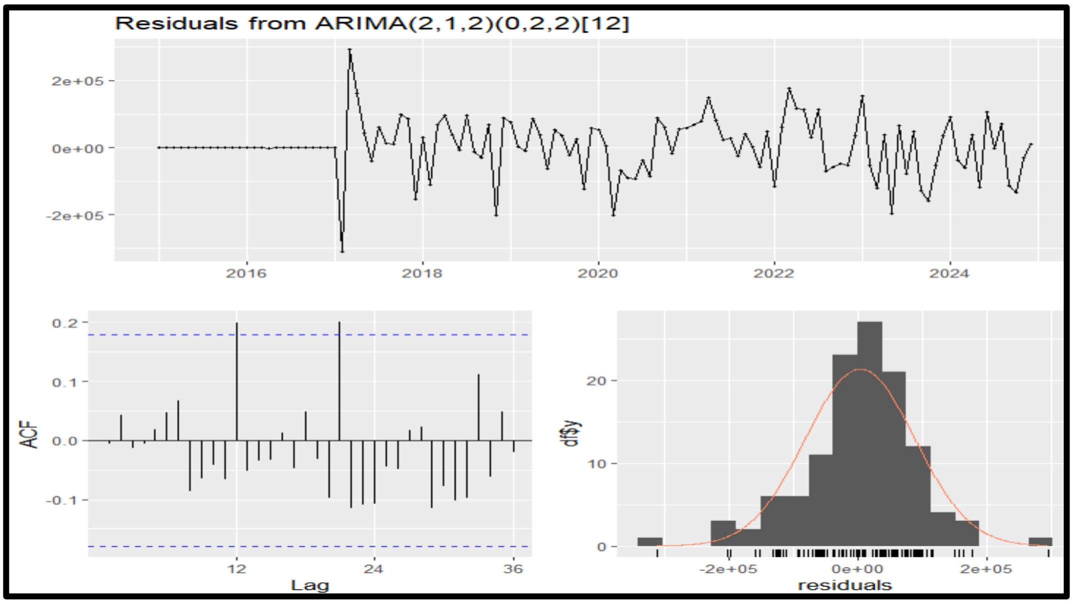
3.6 Model Diagnostics

After fitting the model, residuals were tested to validate model adequacy:

• Ljung-Box test: p-value = **0.1784** (indicating no autocorrelation).

• Runs test: p-value = **0.5839** (indicating randomness in the residuals).

• shapiro-test: p-value= **0.001332** (indicating the residuals does not follow normal)



• ACF and PACF plots showed no significant lags, except for **lag 12**, and **lag 21**, which were tested using the Ljung-Box test. The p-values for these lags were **0.7083** , and **0.662** respectively, indicating no significant autocorrelation at these lags.

Overall, the residual diagnostics indicated that the model was appropriate for forecasting.

3.7 Adjustment of Prediction Intervals to Avoid Negative Bounds

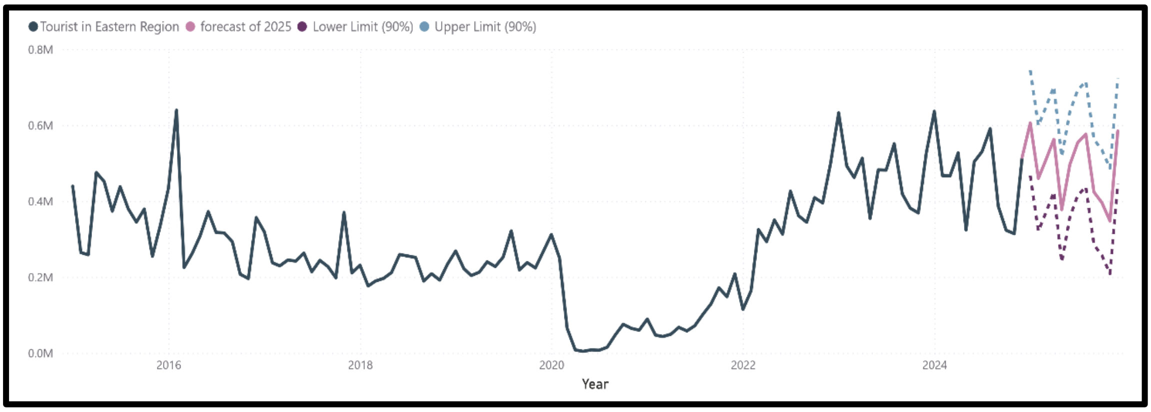
The original 95% prediction intervals generated by the SARIMA(2,1,2)(1,2,0)[12] model included some negative lower bounds, which are not meaningful in the context of tourism data, as the number of tourists cannot be negative. For instance, in certain forecasted months, the lower bounds dropped below zero, indicating a need for adjustment. To address this issue, custom 90% prediction intervals were calculated using the standard deviation of the residuals. This approach ensured all interval values remained positive while maintaining the integrity and interpretability of the forecasts.

3.8 Forecast Results for 2025

The table below presents the forecasted values for each month in 2025, including the lower and upper confidence limits (90% interval):

Month	Forecast	Lower Limit (90%)	Upper Limit (90%)
January	606186.6	467295	745078
February	460131.4	321240	599023
March	507913.1	369021	646805
April	562902.2	424011	701794
May	377835.8	238944	516727
June	496461	357569	635353
July	554333.4	415442	693225
August	576740.6	437849	715632
September	423886.5	284995	562778
October	396554.6	257663	535446
November	347932	209040	486824
December	584779.7	445888	723671

The following chart visualizes the forecasted values for 2025, along with the corresponding confidence intervals. These intervals provide a range within which the actual values are expected to fall, offering a measure of uncertainty in the predictions. The forecast, based on historical data and the applied time series model, aims to provide insights into the expected tourist numbers in 2024, while the confidence limits help in understanding the potential variability and risk associated with the predictions.



3.9 Model Accuracy Evaluation

The forecasting performance of the selected SARIMA(2,1,2)(0,2,2)[12] model was evaluated using the Mean Absolute Percentage Error (MAPE). The model achieved a

MAPE value of **6.78%,** which reflects a high level of accuracy. Generally, a MAPE below 10% is considered to indicate a very good forecasting model, supporting the reliability of the model in capturing the underlying patterns of the data and providing dependable future estimates.

CHAPTER FOUR

CONCLUSION AND REFERNCES

THIS CHAPTER INCLUDES:

- A SUMMARY OF THE KEY FINDINGS FROM THE ANALYSIS.
- THE MAIN CONCLUSIONS DERIVED FROM THE STUDY.
- LIMITATIONS ENCOUNTERED DURING THE RESEARCH PROCESS.
- RECOMMENDATIONS FOR FUTURE STUDIES AND POTENTIAL IMPROVEMENTS.
- FINAL REMARKS ON THE IMPLICATIONS OF THE RESULTS.
- REFERENCES AND SOURCES USED THROUGHOUT THE REPORT.

Conclusion

This study aimed to analyze the number of tourists in Makkah, Riyadh, and the Eastern Region from 2015 to 2023, utilizing SARIMA models to forecast the expected number of tourists for 2024 and 2025. Given the significance of tourism in Saudi Arabia's economy—particularly in Makkah, which hosts millions of pilgrims annually—the ability to predict future trends is crucial for effective planning and policy-making.

By implementing SARIMA modeling with Box-Cox or log transformations, we addressed issues related to non-stationarity and variability in the data. The forecasting model was tested for statistical significance, residual randomness, and model adequacy using Runs Test, Ljung-Box Test, and coefficient significance tests.

Overall, the analysis highlights the seasonal and trend components of tourism in Makkah, Riyadh, and the Eastern Region, providing valuable insights for stakeholders, including policymakers, tourism authorities, and businesses in the hospitality sector.

4.1 Recommendations

Based on the findings of this study, the following recommendations are proposed to enhance tourism forecasting accuracy and improve decision-making processes:

1. Refining the Forecasting Model

• While SARIMA performed reasonably well, alternative approaches such as SARIMAX (including additional external variables), Prophet, or machine learning techniques could be explored in future research.

• Incorporating real-time external factors, such as economic indicators, government policies, or global events, may enhance forecasting accuracy.

• Testing different seasonality structures (e.g., accounting for Islamic calendar-based seasonality) might improve model performance, especially for Hajj and Ramadan periods.

2. Data Enhancement and Collection

• Expanding the dataset by including 2024 actual tourism numbers (once available) will improve the accuracy of future predictions.

• Gathering more granular data, such as monthly or weekly tourist arrivals categorized by nationality or purpose of visit, could lead to more precise modeling.

• Developing a centralized tourism data repository with historical and real-time updates will facilitate better analysis and forecasting.

3. Tourism Strategy and Policy Implications

• The results of this study should be used to aid infrastructure planning, particularly in Makkah, where fluctuations in visitor numbers can significantly impact transportation, accommodation, and public services.

• Policymakers should leverage forecasting insights to optimize resource allocation for peak tourist seasons, ensuring that visitor capacity and services are efficiently managed.

• Given the potential impact of global events like pandemics, it is essential to develop adaptive strategies for crisis management, allowing for dynamic adjustments to tourism policies based on predictive analytics.

4. Integrating Forecasting with Smart Tourism Initiatives

• Utilizing Artificial Intelligence (AI) and Big Data could complement traditional forecasting methods, offering real-time predictive capabilities for tourism management.

• Implementing smart tourism applications that track visitor movements and predict demand surges could enhance crowd management and improve overall visitor experiences.

5. Continuous Monitoring and Model Updates

• As new data becomes available, the forecasting model should be regularly re-evaluated and recalibrated to maintain accuracy.

• Running backtesting analyses using newly acquired data can help assess model performance and identify potential refinements.

• Establishing a collaborative framework between government agencies, research institutions, and private-sector stakeholders can ensure that tourism forecasting remains a dynamic and evolving tool for decision-making.

4.2 Final Thoughts

Accurate tourism forecasting is a vital component of strategic planning in Saudi Arabia, particularly given the Kingdom's Vision 2030 goals for tourism expansion. While SARIMA modeling has provided a strong foundation for this study, there remains significant potential for improving predictive accuracy through enhanced data integration, advanced modeling techniques, and real-time analytics. By adopting a data-driven approach, Saudi Arabia can further strengthen its tourism sector, ensuring sustainable growth and an enhanced visitor experience in the years to come.

4.3 Reference

1. General Authority for Statistics. (2024). Saudi Tourism Data. Retrieved from https://www.stats.gov.sa/en

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